RealTick

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RealTick xAPI Technical Reference Document

This document contains technical details about RealTick xAPI.

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What's New

This release includes the following updates:

Version No.	Date	Summary of Update
v2026.7.0.21	Nov 13, 2026	Added new fields in OrderDetailByDateRangeRequest.



Note: Refer to the Revision History section for detailed information on past release versions.

Introduction

The new Eze EMS xAPI platform is powered by Google's high performance gRPC framework which allows engineers to quickly build efficient and connected systems.

Clients can now seamlessly connect their proprietary applications, custom models and workflows, built in any gRPC supported language - C/C++, Python, Java, Go, or Ruby, to name a few - with Eze EMS to leverage best-in-class global execution capabilities providing comprehensive trading, data, and risk solutions. Eze EMS xAPI is supported on both Windows and Linux.

While language-agnostic, SS&C Eze expects many clients will want to connect their systems to Eze EMS via Python using the gRPC framework. Python has gained immense traction in the quant trading community, in particular, as it makes it easy to build intricate models due to the availability of scientific libraries. Moreover, it is easier to write and evaluate algo trading structures in Python and the code can be extended to dynamic algorithms for trading.

Eze EMS xAPI Basics

The Eze EMS xAPI operates in conjunction with your existing Eze EMS account permissioning and entitlements. **The Eze EMS xAPI is not a standalone data feed application that is provided to you independent of the Eze EMS.** Please contact Eze Client Service if you need to request or make changes to appropriate permissions for your account.

Eze EMS xAPI Use Restrictions

As an Eze EMS xAPI user, you are prohibited from retransmitting any Eze Market Data using the Eze EMS xAPI, without the express prior written consent of Eze EMS and the exchanges or other third-party data providers (referred to as ***"Sources"*** in your end user agreement). Any unauthorized retransmission of Eze Market Data is a breach of your end user agreement and will cause immediate termination of your use of the Eze EMS, Eze Market Data, and the Eze EMS xAPI.

Any non-display usage of Eze Market Data, such as use of real- time data in algorithmic trading or program trading, is subject to the rules, regulations, and policies of the applicable exchanges and additional exchange fees may apply. In addition, you may have a non-display usage of Eze Market Data even if a display of real-time data occurs. Please review your Eze EMS end user agreement, and the exchanges' and third-party data providers' rules, regulations, and policies that apply to your use of the Eze EMS API (which apply to Eze EMS xAPI) and/or Eze Market Data. It is the sole responsibility of the Eze EMS xAPI user and each user receiving, directly or indirectly accessing or otherwise using Eze Market Data to determine whether your receipt, access or use is reportable and/or fee liable.

Pre-Requisites

- The language of choice that the client wants to use. Example: Python version 3.x
- gRPC tools that contains modules such as protobuf and protoc (proto compiler) that are necessary for generating client/ stub files

- gRPC 2.27.0 and above
- Visit this link to learn about language specific pre-requisites: https://grpc.io/docs/languages/

Version

The current stable version of Eze EMS xAPI is 2026.7.0.21.

Endpoint

Contact your SS&C Eze client service representative for more information on Eze EMS xAPI server address with port number.

Resources

The following APIs are provided as part of Eze EMS xAPI:

Order APIs		
API Name	**Input**	**Output**
Order (Single Order APIs)		
CancelSingleOrder	CancelSingleOrderRequest	CancelSingleOrderResponse
ChangeSingleOrder	ChangeSingleOrderRequest	ChangeSingleOrderResponse
SubmitAllocationOrder	SubmitAllocationOrderRequest	SubmitAllocationOrderResponse
SubmitBookTrade	SubmitBookTradeRequest	SubmitBookTradeResponse
SubmitSingleOrder	SubmitSingleOrderRequest	SubmitSingleOrderResponse
SubmitTradeReport	SubmitTradeReportRequest	SubmitTradeReportResponse
Order (Pair Order APIs)		
CancelPairOrder	CancelPairOrderRequest	CancelPairOrderResponse
ChangePairOrder	ChangePairOrderRequest	ChangePairOrderResponse
SubmitPairOrder	SubmitPairOrderRequest	SubmitPairOrderResponse
Order (Basket Order APIs)		
SubmitBasketOrder	BasketOrderRequest	BasketOrderResponse
Order (Miscellaneous APIs)		
GetOrderDetailByDateRange	OrderDetailByDateRangeRequest	stream Order-DetailByDateRangeResponse
GetOrderDetailByOrderId	OrderDetailByOrderIdRequest	OrderDetailByOrderIdResponse
GetOrderDetailByOrderTag	OrderDetailByOrderTagRequest	OrderDetailByOrderTagResponse
GetUserAccounts	UserAccountsRequest	UserAccountsResponse
SubmitSeedData	SubmitSeedDataRequest	SubmitSeedDataResponse
SubscribeOrderInfo	SubscribeOrderInfoRequest	stream SubscribeOrderInfoResponse
SubscribeOrderInfoJson	SubscribeOrderInfoJsonRequest	stream Sub-scribeOrderInfoJsonResponse

MarketData APIs		
API Name	Input	Output
AddSymbols	AddSymbolsRequest	AddSymbolsResponse
GetDailyWeeklyMonthlyBars	DailyWeeklyMonthlyBarsRequest	DailyWeeklyMonthlyBarsResponse
GetDescriptionFromOptionSymbol	DescriptionFromOptionSymbolRequest	DescriptionFromOptionSymbolResponse
GetIntradayBars	IntradayBarsRequest	IntradayBarsResponse
GetLevel1MarketData	Level1MarketDataRecordRequest	Level1MarketDataRecordResponse
GetOptionChainForUnderlier	OptionChainRequest	OptionChainResponse
GetOptionsAndGreekData	OptionsAndGreekDataRequest	OptionsAndGreekDataResponse
GetOptionSymbolFromDescription	OptionSymbolFromDescriptionRequest	OptionSymbolFromDescriptionResponse
GetSecurityData	SecurityDataRequest	SecurityDataResponse
GetSymbolFromAlternateSymbology	SymbolFromAlternateSymbologyRequest	SymbolFromAlternateSymbologyResponse
GetSymbolReferenceData	SymbolReferenceDataRequest	SymbolReferenceDataResponse
GetSymbolsFromCompanyName	SymbolsFromCompanyNameRequest	SymbolsFromCompanyNameResponse
GetTickData	TickDataRequest	TickDataResponse
RemoveSymbols	RemoveSymbolsRequest	RemoveSymbolsResponse
SubscribeIntradayBars	SubscribeIntradayBarsRequest	stream SubscribeIntradayBarsResponse
StreamMarketData	MarketDataStreamRequest	MarketDataStreamResponse
SubscribeLevel1Ticks	Level1MarketDataRequest	stream Level1MarketDataResponse
SubscribeLevel2Ticks	Level2MarketDataRequest	stream Level2MarketDataResponse
SubscribeTickData	SubscribeTickDataRequest	stream SubscribeTickDataResponse
UnSubscribeIntradayBars	UnSubscribeIntradayBarsRequest	UnSubscribeIntradayBarsResponse
UnSubscribeLevel1Data	UnSubscribeLevel1DataRequest	UnSubscribeLevel1DataResponse

MarketData APIs		
API Name	Input	Output
UnSubscribeLevel2Data	UnSubscribeLevel2DataRequest	UnSubscribeLevel2DataResponse
UnSubscribeStreamMarketData	UnSub-scribeStreamMarketDataRequest	UnSub-scribeStreamMarketDataResponse
UnSubscribeTickData	UnSubscribeTickDataRequest	UnSubscribeTickDataResponse

Utility APIs		
API Name	Input	Output
ChangePassword	ChangePasswordRequest	ChangePasswordResponse
Connect	ConnectRequest	ConnectResponse
CreateStrategy	CreateStrategyRequest	CreateStrategyResponse
Disconnect	DisconnectRequest	DisconnectResponse
GetFirmName	GetFirmNameRequest	GetFirmNameResponse
GetStrategyData	GetStrategyDataRequest	GetStrategyDataResponse
GetStrategyList	StrategyListRequest	StrategyListResponse
GetTodaysActivity	TodaysActivityRequest	TodaysActivityResponse
GetTodaysActivityJson	TodaysActivityJsonRequest	TodaysActivityJsonResponse
GetTodaysBalances	TodaysBalancesRequest	TodaysBalancesResponse
GetTodaysBrokenDownPos-itions	TodaysBrokenDownPos-itionsRequest	TodaysBrokenDownPos-itionsResponse
GetTodaysNetPositions	TodaysNetPositionsRequest	TodaysNetPositionsResponse
GetUserRouteProps	GetUserRoutePropsRequest	GetUserRoutePropsResponse
GetUserRoutes	GetUserRoutesRequest	GetUserRoutesResponse
MultiFactorAuthentication	MultiFactorAuthenticationRequest	MultiFactorAuthenticationResponse
SubscribeHeartBeat	SubscribeHeartBeatRequest	SubscribeHeartBeatResponse
SubscribeTodaysNetPositions	SubscribeNetPositionsRequest	stream Sub-scribeNetPositionsResponse

Utility APIs		
API Name	Input	Output
UnSubscribeTodaysNetPositions	UnSubscribeNetPositionsRequest	UnSubscribeNetPositionsResponse
UpdateStrategy	UpdateStrategyRequest	UpdateStrategyResponse
UserExists	UserExistsRequest	UserExistsResponse
Secured Remote Password APIs		
ChangePasswordSRP	ChangePasswordSRPRequest	ChangePasswordSRPResponse
CompleteLoginSrp	CompleteLoginSrpRequest	CompleteLoginSrpResponse
StartLoginSrp	StartLoginSrpRequest	StartLoginSrpResponse

Note: Only the mandatory API fields are provided in this document. Contact your SS&C Eze client service representative for a complete list of extended fields.

Note: Ready-to-use sample scripts for the APIs described in this document are available in Python, Java, .NET, and Go on the project Git repository. Refer to the corresponding language-specific package in the repository for implementation examples and usage guidance.

Authentication APIs

Connect

The **ConnectRequest** object is used as an input in **Connect** API. This API is used to login to EMS xAPI to connect to the server.

ConnectRequest

Field Name	Required?	Data Type	Accepted Values/Examples
Domain	Yes	string	A valid domain name
Locale	Yes	string	The region the user wants to connect (e.g., AMERICAS, ASIA)
Password	Yes	string	A valid password
UserName	Yes	string	A valid user name

Note: Your account will be temporarily locked for 3 minutes if you attempt to log in three (3) times within 60 seconds, regardless of whether the attempts are successful or unsuccessful.

ConnectResponse

Field Name	Data Type	Retrieves
Acknowledgement	ServerAcknowledgement	Response from the server (success or failure). Refer to the section ServerAcknowledgement
DPermStatusCode	string	The dynamic permission status code associated with the connected session
MFASecurityCode	string	The MFA security code issued upon a successful connection, to be used for subsequent multi-factor authentication verification
MFAtotpPreference	string	The user's preferred Time-based One-Time Password (TOTP) method configured for multi-factor authentication
MFAtotpValue	string	Time-based One-Time Password (TOTP) value associated with the user's configured MFA authentication method
Response	string	A valid user token

Field Name	Data Type	Retrieves
UserToken	string	A server generated GUID that is given as response to the client during the first login

Disconnect

The **DisconnectRequest** object is used as an input in **Disconnect** API. This API is used to disconnect EMS xAPI server and end the session.

DisconnectRequest

Field Name	Required?	Data Type	Accepted Values/Examples
UserToken	Yes	string	A server generated GUID that is given as response to the client during the first login
TimeoutInSeconds	No	int32	Time out interval for the API request

DisconnectResponse

Field Name	Data Type	Retrieves
Acknowledgement	ServerAcknowledgement	Response from the server (success or failure). Refer to the section ServerAcknowledgement
OptionalFields	map<string, string>	This is additional information given to the client in cases of failure. The field OptionalFields ["ErrorMessage"] would contain the reason for error or exception
ServerResponse	string	Success: In case the invoked API executes successfully on the server side Failed: In case the invoked API throws an error or lands in an exception

ChangePassword

The **ChangePasswordRequest** object is used as an input in **ChangePassword** API. You can use this API to change your password. The updated password is applicable to both EMS xAPI and EMS.

ChangePasswordRequest

Field Name	Required?	Data Type	Accepted Values/Examples
Domain	Yes	string	A valid domain name

Field Name	Required?	Data Type	Accepted Values/Examples
NewPassword	Yes	string	Enter new password
OldPassword	Yes	string	Enter current password
UserName	Yes	string	A valid user name

ChangePasswordResponse

Field Name	Data Type	Retrieves
Acknowledgement	ServerAcknowledgement	Response from the server (success or failure). Refer to the section ServerAcknowledgement
message	string	Server response to change password SRP API
status	string	A server generated GUID that is given as response to the client during the first login

MultiFactorAuthentication

The **MultiFactorAuthenticationRequest** object is used as input to the **MultiFactorAuthentication** API. Use this API to complete multi-factor authentication when notified in the **Connect** API.

1. Call the Connect API with your credentials.
2. If multi-factor authentication is required, the ConnectResponse returns a MFASecurityCode. A VerificationCode is also sent to you via the configured authentication method, such as email or SMS.
3. Call the MultiFactorAuthentication API with the MFASecurityCode and the VerificationCode.
4. Upon successful authentication, the API returns a UserToken for use in subsequent requests.

MultiFactorAuthenticationRequest

Field Name	Required?	Data Type	Accepted Values/Examples
Domain	Yes	string	A valid domain name
Locale	Yes	string	The region the user wants to connect (e.g., AMERICAS, ASIA)
MFASecurityCode	Yes	string	A ticket returned by the Connect API that identifies the pending multi-factor authentication request and must be provided when completing authentication

Field Name	Required?	Data Type	Accepted Values/Examples
UserName	Yes	string	A valid user name
VerificationCode	Yes	string	The verification code sent to the user via email, SMS, or another configured multi-factor authentication method

MultiFactorAuthenticationResponse

Field Name	Data Type	Retrieves
Acknowledgement	ServerAcknowledgement	Response from the server (success or failure). Refer to the section ServerAcknowledgement
Response	string	Success: In case the invoked API executes successfully on the server side Failed: In case the invoked API throws an error or lands in an exception
UserToken	string	A server generated GUID that is given as response to the client during the first login

Secured Remote Password

Secured Remote Password (SRP) is a mechanism using which a client can encrypt sensitive information such as password by making use of public and private keys. By implementing the SRP support in EMS xAPI server, the clients can be equipped to send hash coded password during login.

StartLoginSrp

The **StartLoginSrpRequest** object is used as an input in **StartLoginSrp** API. This API is used to login to EMS xAPI using the SRP method which utilizes hash coded strings to exchange the password.

StartLoginSrpRequest

Field Name	Required?	Data Type	Accepted Values/Examples
Domain	Yes	string	A valid domain name
UserName	Yes	string	A valid user name

Note: Your account will be temporarily locked for 3 minutes if you attempt to log in three (3) times within 60 seconds, regardless of whether the attempts are successful or unsuccessful.

StartLoginSrpResponse

Field Name	Data Type	Retrieves
Acknowledgement	ServerAcknowledgement	Response from the server (success or failure). Refer to the section ServerAcknowledgement
Response	string	Success: In case the invoked API executes successfully on the server side Failed: In case the invoked API throws an error or lands in an exception
srpb	string	This field is specific to SRP6 Protocol
srpg	string	This field is specific to SRP6 Protocol
srpN	string	This field is specific to SRP6 Protocol
srpSalt	string	This field is specific to SRP6 Protocol
srpTransactId	string	Unique transaction ID per user
Status	string	Success or Failure of API call

Note: StartLoginSrp has to be followed by a call to **CompleteLoginSrp**. The output from **StartLoginSrp** is used as a direct/indirect input to **CompleteLoginSrp**.

CompleteLoginSrp

The **CompleteLoginSrpRequest** object is used as an input in **CompleteLoginSrp** API. This API is used to complete login to SRP using your identity, transactionId, strEphA and strMc parameters.

CompleteLoginSrpRequest

Field Name	Required?	Data Type	Accepted Values/Examples
Domain	Yes	string	A valid domain name
Identity	Yes	string	A valid user identity (combination of user-name@domain)
Locale	Yes	string	The region the user wants to connect (e.g., AMERICAS, ASIA)
srpTransactId	Yes	string	Unique transaction ID per user
strEphA	Yes	string	This field is specific to SRP6 Protocol
strMc	Yes	string	This field is specific to SRP6 Protocol

Field Name	Required?	Data Type	Accepted Values/Examples
UserName	Yes	string	A valid user name

CompleteLoginSrpResponse

Field Name	Data Type	Retrieves
Acknowledgement	ServerAcknowledgement	Response from the server (success or failure). Refer to the section ServerAcknowledgement
Response	string	Success: In case the invoked API executes successfully on the server side Failed: In case the invoked API throws an error or lands in an exception
UserToken	string	A server generated GUID that is given as response to the client during the first login

ChangePasswordSRP

The **ChangePasswordSRPRequest** object is used as an input in **ChangePasswordSRP** API. You can use this API to change your password. The updated password is applicable to both EMS xAPI and EMS.

ChangePasswordSRPRequest

Field Name	Required?	Data Type	Accepted Values/Examples
Domain	Yes	string	A valid domain name
NewPassword	Yes	string	Enter new password
OldPassword	Yes	string	Enter current password
TransactId	Yes	string	A server generated GUID that is given as response to the client during the first login
UserName	Yes	string	A valid user name

ChangePasswordSRPResponse

Field Name	Data Type	Retrieves
Acknowledgement	ServerAcknowledgement	Response from the server (success or failure). Refer to the section ServerAcknowledgement
message	string	Server response to change password SRP API

Field Name	Data Type	Retrieves
status	string	A server generated GUID that is given as response to the client during the first login

MarketData APIs

The following APIs are provided as part of MarketData:

API Name	Input	Output
AddSymbols	AddSymbolsRequest	AddSymbolsResponse
GetDailyWeeklyMonthlyBars	DailyWeeklyMonthlyBarsRequest	DailyWeeklyMonthlyBarsResponse
GetDescriptionFromOptionSymbol	DescriptionFromOptionSymbolRequest	DescriptionFromOptionSymbolResponse
GetIntradayBars	IntradayBarsRequest	IntradayBarsResponse
GetLevel1MarketData	Level1MarketDataRecordRequest	Level1MarketDataRecordResponse
GetOptionChainForUnderlier	OptionChainRequest	OptionChainResponse
GetOptionsAndGreekData	OptionsAndGreekDataRequest	OptionsAndGreekDataResponse
GetOptionSymbolFromDescription	OptionSymbolFromDescriptionRequest	OptionSymbolFromDescriptionResponse
GetSecurityData	SecurityDataRequest	SecurityDataResponse
GetSymbolFromAlternateSymbology	SymbolFromAlternateSymbologyRequest	SymbolFromAlternateSymbologyResponse
GetSymbolReferenceData	SymbolReferenceDataRequest	SymbolReferenceDataResponse
GetSymbolsFromCompanyName	SymbolsFromCompanyNameRequest	SymbolsFromCompanyNameResponse
GetTickData	TickDataRequest	TickDataResponse
RemoveSymbols	RemoveSymbolsRequest	RemoveSymbolsResponse
SubscribeIntradayBars	SubscribeIntradayBarsRequest	stream SubscribeIntradayBarsResponse
StreamMarketData	MarketDataStreamRequest	MarketDataStreamResponse
SubscribeLevel1Ticks	Level1MarketDataRequest	stream Level1MarketDataResponse

API Name	Input	Output
SubscribeLevel2Ticks	Level2MarketDataRequest	stream Level2MarketDataResponse
SubscribeTickData	SubscribeTickDataRequest	stream SubscribeTickDataResponse
UnSubscribeIntradayBars	UnSubscribeIntradayBarsRequest	UnSubscribeIntradayBarsResponse
UnSubscribeLevel1Data	UnSubscribeLevel1DataRequest	UnSubscribeLevel1DataResponse
UnSubscribeLevel2Data	UnSubscribeLevel2DataRequest	UnSubscribeLevel2DataResponse
UnSubscribeStreamMarketData	UnSubscribeStreamMarketDataRequest	UnSubscribeStreamMarketDataResponse
UnSubscribeTickData	UnSubscribeTickDataRequest	UnSubscribeTickDataResponse

Custom Data Type

AlternateSymbology

Field Name	Data Type	Accepted Values/Examples
Symbology	enum	ISIN = 0; SEDOL = 1; RIC = 2; CUSIP = 3; BBG = 4;
SymbolOption	Symbology	Option series symbol 1

CommonBarsFields

Field Name	Data Type	Accepted Values/Examples
AcVol1	repeated int32	Total volume
High1	repeated Price	Value of highest transaction for current session. Refer to the section Price
Low1	repeated Price	Value of lowest transaction for current session. Refer to the section Price
OpenPrc	repeated Price	Open price of current session. Refer to the section Price

Field Name	Data Type	Accepted Values/Examples
Settle	repeated Price	Last settlement price. Refer to the section Price

Interval

Field Name	Data Type	Accepted Values/Examples
Options	enum	DAILY = 0; WEEKLY = 1; MONTHLY = 2

IntradayBarUpdate

Field Name	Data Type	Accepted Values/Examples
AcVol1	int32	Total volume
High1	Price	Value of highest transaction for current session. Refer to the section Price
Kind	BarUpdateKind	BAR_KIND_UNSPECIFIED = 0 BAR_KIND_HISTORICAL = 1 BAR_KIND_PARTIAL = 2 BAR_KIND_CLOSED = 3
Low1	Price	Value of lowest transaction for current session. Refer to the section Price
OpenPrc	Price	Open price of current session. Refer to the section Price
OpenTime	google.protobuf.Timestamp	Opening time of the intraday bar interval
Settle	Price	Last settlement price. Refer to the section Price
Symbol	string	Valid ticker symbol (e.g., AAPL, IBM, VOD.LSE)
TickCount	int32	Number of ticks (trades) occurred within the intraday bar interval
TrdTim1	google.protobuf.Duration	Trade time
Vwap	google.protobuf.DoubleValue	VWAP from the trade server

Level1FieldOptions

Field Name	Data Type	Accepted Values/Examples
Fields	enum	DISP_NAME = 0 TRDPRC1 = 1 TRDTIM1 = 2 SYMBOL_DESC = 3 COMPANY_NAME = 4 ARCA_IMBALANCE_VOLUME = 5 ARCA_MATCH_VOLUME = 6 SALE_CONDITION_VOLUME = 7 INTRADAY_HIGH_COUNT = 8 VWAP_VOL = 9 VWAP = 10 BID = 11 ASK = 12 CHANGE_LAST = 13 HIGH1 = 14 HIGH52 = 15 LOW1 = 16 LOW52 = 17 EXTENDED_FIELDS = 18
RequestedExtendedFields	repeated string	Fetch details for the specified extended fields
RequestedFields	repeated Fields	Fetch details for the specified fields

The **Level1MarketDataRequest** object is used as an input in **GetLevel1MarketData** API. This is an unary API used to fetch level 1 market data snapshot (e.g., Bid, Ask, Last traded price). You can request one or more symbols data at a time.

Level1MarketDataRecord

Field Name	Data Type	Accepted Values/Examples
ArcaImbalanceVolume	google.protobuf.Int32Value	Pre-Open Imbalance Volume (positive is Bid, negative is Offer)
ArcaMatchVolume	google.protobuf.Int32Value	Total number of shares to be executed
Ask	Price	Last ask price for symbol
Bid	Price	Last bid price for symbol
BidTick2	string	"Bid tick" for NASDAQ Short Sale Rule

Field Name	Data Type	Accepted Values/Examples
ChangeLast	Price	Change from last price
CompanyName	string	Company name represented by symbol
DispName	string	'IBM' (Symbol formats are datafeed-specific. Check your Feed Handler help file for the correct format.)
High1	Price	Value of highest transaction for current session
High52	Price	Highest traded price for a share in the last 52 weeks
IntradayHighCount	google.protobuf.Int32Value	Number of intraday highs
Low1	Price	Value of lowest transaction for current session
Low52	Price	Lowest traded price for a share in the last 52 weeks
LrpAsk	Price	NYSE liquidity replenishment point for Asks
LrpBid	Price	NYSE liquidity replenishment point for bids
SaleConditionVolume	google.protobuf.Int32Value	Volume of the most recent pre-market, post-market, or irregular trade
SymbolDesc	string	Symbol description
Trdprc1	Price	Last traded price
Trdtim1	google.protobuf.Duration	Trade time
Vwap	google.protobuf.DoubleValue	VWAP from the trade server
VwapVol	google.protobuf.Int32Value	VWAP Volume from the trade server

OptionCalculationResponse

Field Name	Data Type	Retrieves
Delta	google.protobuf.DoubleValue	The ratio which compares the change in price of an underlying asset to the corresponding change in the price of a derivative
DispName	string	'IBM' (Symbol formats are datafeed-specific. Check your feed handler help file for the correct format.)
DividendAmout1	google.protobuf.DoubleValue	1st dividend amount
DividendAmout2	google.protobuf.DoubleValue	2nd dividend amount
DividendAmout3	google.protobuf.DoubleValue	3rd dividend amount
DividendAmout4	google.protobuf.DoubleValue	4th dividend amount
DividendAmout5	google.protobuf.DoubleValue	5th dividend amount
DividendAmout6	google.protobuf.DoubleValue	6th dividend amount
DividendDate1	google.protobuf.Timestamp	1st dividend date
DividendDate2	google.protobuf.Timestamp	2nd dividend date
DividendDate3	google.protobuf.Timestamp	3rd dividend date
DividendDate4	google.protobuf.Timestamp	4th dividend date
DividendDate5	google.protobuf.Timestamp	5th dividend date
DividendDate6	google.protobuf.Timestamp	6th dividend date
Gamma	google.protobuf.DoubleValue	Rate of change for delta with respect to the underlying asset's price
ImpliedVol	google.protobuf.DoubleValue	The estimated volatility of a security's price
Model	google.protobuf.Int32Value	Option model used in the calculation
OptionRoot	string	Option root for all exiration months and instrument types
OriginalPrice	Price	Original price
Premium	google.protobuf.DoubleValue	Displays a calculated theoretical value for an option or warrant

Field Name	Data Type	Retrieves
PutCallId	string	Put/Call ID
Rate	google.protobuf.DoubleValue	Interest rate used in the calculation
StrikePrice	Price	Price at which option can be exercised
Theta	google.protobuf.DoubleValue	A measure of a rate of decline in the value of an option due to the passage of time
TimeToExpire	google.protobuf.DoubleValue	Time to expiration
UnderlierPrice	Price	Underlier price at time of order placement
UnderSym	string	Underlier symbol
Vega	google.protobuf.DoubleValue	The amount that the price of an option changes compared to a 1% change in volatility

OptionTypes

Field Name	Data Type	Accepted Values/Examples
Options	enum	CALL = 0; PUT = 1

QuoteChainResponse

Field Name	Data Type	Retrieves
Basisvalue	google.protobuf.DoubleValue	Currency value of a unit move
CommodityName	string	Name of the contract for a commodity
DispName	string	'IBM' (Symbol formats are datafeed-specific. Check your feed handler help file for the correct format.)
ExchName	string	Three-letter acronym for listing exchange
ExpirDate	google.protobuf.Timestamp	Date of expiration for an option
Minmove	google.protobuf.Int32Value	Minimum movement
PutCallInd	string	Option type (P = put, C = call, U = underlier) for symbol

Field Name	Data Type	Retrieves
Session	google.protobuf.Int32Value	Session indicator (specified by data feed)
StrikePrice	Price	Price at which option can be exercised
Styp	google.protobuf.Int32Value	Security type
SymbolDesc	string	Symbol description
TrdUnits	string	Units for expressing prices for this symbol

SecurityData

Field Name	Data Type	Accepted Values/Examples
Adx14d1d	google.protobuf.DoubleValue	ADX 14d
Beta	Price	Calculated using data from the S&P 500 portfolio for 12 months prior to the current month
BloombergCode	string	Bloomberg symbol code associated with order
BloombergCodeComposite	string	Bloomberg code composite
BollingerLower21d1d	google.protobuf.DoubleValue	Bollinger lower 21d
BollingerUpper21d1d	google.protobuf.DoubleValue	21-day upper envelope
Close5davg1d	google.protobuf.DoubleValue	5-day average price
Close10davg1d	google.protobuf.DoubleValue	10-day average price
Close20davg1d	google.protobuf.DoubleValue	20-day average price
Close50davg1d	google.protobuf.DoubleValue	50-day average price
Close200davg1d	google.protobuf.DoubleValue	200-day average price
CompanyName	string	Company name represented by symbol
Country	string	Country of exchange where stock is listed
Cusip	string	ID assigned to all North American stocks and registered bonds

Field Name	Data Type	Accepted Values/Examples
DispName	string	'IBM' (Symbol formats are datafeed-specific. Check your feed handler help file for the correct format.)
Dividend	Price	Last reported dividend to be paid to shareholders (per share)
DividendFreqN	google.protobuf.Int32Value	Dividend frequency (per year)
DividendRate	Price	Indicated annual dividend rate based on latest payment
DividendYield	Price	Dividend paid per share vs. share price
Divpaydate	google.protobuf.Timestamp	Date on which dividend is paid
Earnings	Price	Net income for the company represented by the selected symbol
Exdivdate	google.protobuf.Timestamp	Ex-div date
GicsIndustry	string	Industry code as specified by Global Industry Classification Standard (GICS)
GicsSector	string	Sector code as specified by Global Industry Classification Standard (GICS)
GicsSubindustry	string	Sub-industry code as specified by Global Industry Classification Standard (GICS)
High15dmax1d	google.protobuf.DoubleValue	15-day high
High52	Price	Highest traded price for a share in the last 52 weeks
High52Date	google.protobuf.Timestamp	Date on which 52-week high was reached
Hlvolatility10d1d	google.protobuf.DoubleValue	10-day high/low volatility

Field Name	Data Type	Accepted Values/Examples
IsinNo	string	International ID assigned to all securities foreign or domestic
IssuersSectorStr	string	Indicates the market sector to which the security belongs, in relation to type
Low15dmin1d	google.protobuf.DoubleValue	15-day low
Low52	Price	Lowest traded price for a share in the last 52 weeks
Low52Date	google.protobuf.Timestamp	Date on which 52-week low was reached
MgSicm	Price	Short interest for current month
MIC	string	Segmented market identification code
Minusdi14d1d	google.protobuf.DoubleValue	14-day minus directional indicator
MktCap	Price	Market capitalization
Peratio	Price	A ratio which calculates current share price vs. per-share earnings
Plusdi14d1d	google.protobuf.DoubleValue	14-day plus directional indicator
PrimaryExchange	string	Primary exchange
ProcDate	google.protobuf.Timestamp	Process date
Region	string	Region associated with the listed exchanges for a symbol
RicCode	string	Reuters instrument code
Rsi9d1d	google.protobuf.DoubleValue	Value of 9-day relative strength index
Rsi14d1d	google.protobuf.DoubleValue	Value of 14-day relative strength index
Rsi25d1d	google.protobuf.DoubleValue	Value of 25-day relative strength index

Field Name	Data Type	Accepted Values/Examples
SecurityCategory	google.protobuf.Int32Value	SIC code
Sedol	string	(Stock Exchange Daily Official List) 7-digit ID assigned to all securities trading on LSE and other smaller UK exchanges
SharesOut	google.protobuf.Int32Value	Shares outstanding
SplitDate1	google.protobuf.Timestamp	Date of first split
SplitFactor1	google.protobuf.DoubleValue	Factor of most recent split for symbol
Volume5davg1d	google.protobuf.DoubleValue	5-day average volume
Volume10davg1d	google.protobuf.DoubleValue	10-day average volume
Volume20davg1d	google.protobuf.DoubleValue	20-day average volume
Volume50davg1d	google.protobuf.DoubleValue	50-day average volume
Volume200davg1d	google.protobuf.DoubleValue	200-day average volume

SymbolData

Field Name	Data Type	Accepted Values/Examples
BloombergCode	string	Bloomberg symbol code associated with order
BloombergCodeComposite	string	Bloomberg code composite
BloombergCodeFull	string	Full bloomberg code
CommodityName	string	Name of the contract for a commodity
Country	string	Country of exchange where stock is listed
Cusip	string	ID assigned to all North American stocks and registered bonds

Field Name	Data Type	Accepted Values/Examples
DispName	string	'IBM' (Symbol formats are datafeed-specific. Check your feed handler help file for the correct format.)
ExchName	string	Three-letter acronym for listing exchange
GicsIndustry	string	Industry code as specified by Global Industry Classification Standard (GICS)
GicsSector	string	Sector code as specified by Global Industry Classification Standard (GICS)
GicsSubindustry	string	Sub-industry code as specified by Global Industry Classification Standard (GICS)
IsinNo	string	International ID assigned to all securities foreign or domestic
RicCode	string	Reuters instrument code
Sedol	string	(Stock Exchange Daily Official List) 7-digit ID assigned to all securities trading on LSE and other smaller UK exchanges
Styp	google.protobuf.Int32Value	Security type
SymbolDesc	string	Symbol description

SymbolLookupResult

Field Name	Data Type	Accepted Values/Examples
Acknowledgement	ServerAcknowledgement	Response from the server (success or failure). Refer to the section ServerAcknowledgement
RequestedSymbol	string	Requested symbol
SymbolInfolist	repeated SymbolData	Refer to the section SymbolData

SymInfoResponse

Field Name	Data Type	Retrieves
BloombergCode	string	Bloomberg symbol code associated with order
BloombergCodeComposite	string	Bloomberg code composite
BloombergCodeFull	string	Full bloomberg code
DispName	string	'IBM' (Symbol formats are datafeed-specific. Check your feed handler help file for the correct format.)
ExchName	string	Three-letter acronym for listing exchange
Styp	google.protobuf.Int32Value	Security type
SymbolDesc	string	Symbol description

TicksResponse

Field Name	Data Type	Retrieves
DispName	repeated string	'IBM' (Symbol formats are datafeed-specific. Check your feed handler help file for the correct format.)
TickType	repeated int32	Tick type
TrdDate	repeated google.-protobuf.Timestamp	Date of last trade
TrdPrc1	repeated Price	Last traded price
TrdTim1	repeated google.-protobuf.Duration	Time of last trade
TrdVol1	repeated int32	Volume of last reported trade
TrdXid1	repeated string	Exchange ID of last trade

TickTypes

Field Name	Data Type	Accepted Values/Examples
Ticks	enum	TRADE = 0; BID = 1; ASK = 2; REGIONAL_BID = 3; REGIONAL_ASK = 4; DELETED = 11; INSERTED = 12; IRREGULAR_DELETE = 44; FORM_T_TRADE = 32

AddSymbols

The **AddSymbolsRequest** object is used as an input in **AddSymbols** API. This API is used to add the symbols to your account.

AddSymbolsRequest

Field Name	Data Type	Accepted Values/Examples
MarketDataLevel	string	Level1 or Level2
Symbols	repeated string	Valid ticker symbols
UserToken	string	A server generated GUID that is given as response to the client during the first login

AddSymbolsResponse

Field Name	Data Type	Retrieves
Acknowledgement	ServerAcknowledgement	Response from the server (success or failure). Refer to the section ServerAcknowledgement
OptionalFields	map<string, string>	This is additional information given to the client in cases of failure. The field OptionalFields ["ErrorMessage"] would contain the reason for error or exception
ServerResponse	string	Success: In case the invoked API executes successfully on the server side Failed: In case the invoked API throws an error or lands in an exception

GetDailyWeeklyMonthlyBars

The **DailyWeeklyMonthlyBarsRequest** object is used as an input in **GetDailyWeeklyMonthlyBars** API. This API is used to request for daily, weekly and monthly bars.

DailyWeeklyMonthlyBarsRequest

Field Name	Data Type	Accepted Values/Examples
DaysBack	google.protobuf.Int32Value	The number of days worth of bars to retrieve
Interim	Interval	Refer to the section Interval
RequestId	google.protobuf.Int32Value	A user-supplied request ID that can be used to dis-ambiguate response data
StopDate	google.protobuf.Timestamp	Stop date (LME - used with start date only)
Symbol	string	Valid ticker symbol (e.g., AAPL, IBM, VOD.LSE)
UserToken	string	A server generated GUID that is given as response to the client during the first login

DailyWeeklyMonthlyBarsResponse

Field Name	Data Type	Retrieves
Acknowledgement	ServerAcknowledgement	Response from the server (success or failure). Refer to the section ServerAcknowledgement
BarFields	CommonBarsFields	Refer to the section CommonBarsFields
DispName	string	'IBM' (Symbol formats are datafeed-specific. Check your Feed Handler help file for the cor-rect format.)

GetDescriptionFromOptionSymbol

The **DescriptionFromOptionSymbolRequest** object is used as an input in **GetDescriptionFromOptionSymbol** API. This API is used to request description from option symbol.

DescriptionFromOptionSymbolRequest

Field Name	Data Type	Accepted Values/Examples
Symbol	string	Valid ticker symbol (e.g., AAPL, IBM, VOD.LSE)
UserToken	string	A server generated GUID that is given as response to the client during the first login

DescriptionFromOptionSymbolResponse

Field Name	Data Type	Retrieves
Acknowledgement	ServerAcknowledgement	Response from the server (success or failure). Refer to the section ServerAcknowledgement
Expiration	string	Time at which order is no longer valid (DAY, DAYPLUS, or custom value supported by destination)
OptionType	OptionTypes	Refer to the section OptionTypes
Root	string	Symbol root for option
StrikePrice	double	Price at which option can be exercised

GetIntradayBars

The **IntradayBarsRequest** object is used as an input in **GetIntradayBars** API. This API is used to acquire intraday bars.

IntradayBarsRequest

Field Name	Data Type	Accepted Values/Examples
BarInterval	int32	Bar interval in minutes
Date	google.protobuf.Timestamp	The end date for the query, i.e. the most recent date for which bars should be retrieved. If null, defaults to the current date
DaysBack	int32	The number of days worth of bars to retrieve
requestId	google.protobuf.Int32Value	A user-supplied request ID that can be used to disambiguate response data
ShowPrePostMarket	bool	If set to **True**, displays the pre/post trading hours market data
StartAtMidnight	bool	If set to **True**, the first bar of the day includes all data since midnight, otherwise the first bar of the day includes all data since the session open time

Field Name	Data Type	Accepted Values/Examples
StartTime	google.protobuf.Duration	Bars earlier than this time will be excluded. If null, there is no restriction
StopTime	google.protobuf.Duration	Bars later than this time will be excluded. If null, there is no restriction
Symbol	string	Valid ticker symbol (e.g., AAPL, IBM, VOD.LSE)
UserToken	string	A server generated GUID that is given as response to the client during the first login

IntradayBarsResponse

Field Name	Data Type	Retrieves
Acknowledgement	ServerAcknowledgement	Response from the server (success or failure). Refer to the section ServerAcknowledgement
Bars	CommonBarsFields	Refer to the section CommonBarsFields
TrdTim1	repeated string	Trade time

GetLevel1MarketData

The **Level1MarketDataRecordRequest** object is used as an input in **GetLevel1MarketData** API. This is an unary API used to fetch level 1 market data snapshot (e.g., Bid, Ask, Last traded price). You can request one or more symbols data at a time.

Level1MarketDataRecordRequest

Field Name	Data Type	Accepted Values/Examples
FieldSelection	optional Level1FieldOptions	Field selection
RegionalExchangeIds	repeated string	Regional exchange ID
Symbols	repeated string	Valid ticker symbol (e.g., AAPL, IBM, VOD.LSE)
UserToken	string	A server generated GUID that is given as response to the client during the first login

Level1MarketDataRecordResponse

Field Name	Data Type	Retrieves
Acknowledgement	ServerAcknowledgement	Response from the server (success or failure). Refer to the section ServerAcknowledgement
DataRecord	repeated Level1MarketDataRecord	Refer to the section Level1MarketDataRecord

GetOptionChainForUnderlier

The **OptionChainRequest** object is used as an input in **GetOptionChainForUnderlier** API. This API is used to acquire the option chain.

OptionChainRequest

Field Name	Data Type	Accepted Values/Examples
RequestId	google.protobuf.Int32Value	A user-supplied request ID that can be used to disambiguate response data
SymbolRoot	string	Symbol root for options
UserToken	string	A server generated GUID that is given as response to the client during the first login

OptionChainResponse

Field Name	Data Type	Retrieves
Acknowledgement	ServerAcknowledgement	Response from the server (success or failure). Refer to the section ServerAcknowledgement
Derivative	repeated QuoteChainResponse	Underlier information

GetOptionsAndGreekData

The **OptionsAndGreekDataRequest** object is used as an input in **GetOptionsAndGreekData** API. This API is used to request the options and greek data.

OptionsAndGreekDataRequest

Field Name	Data Type	Accepted Values/Examples
Symbols	repeated string	Valid ticker symbol (e.g., AAPL, IBM, VOD.LSE)
UserToken	string	A server generated GUID that is given as response to the client during the first login

OptionsAndGreekDataResponse

Field Name	Data Type	Retrieves
Acknowledgement	ServerAcknowledgement	Response from the server (success or failure). Refer to the section ServerAcknowledgement
OptionsList	repeated OptionCalculationResponse	Refer to the section OptionCalculationResponse

GetOptionSymbolFromDescription

The **OptionSymbolFromDescriptionRequest** object is used as an input in **GetOptionSymbolFromDescription** API. This API is used to request option symbol from description.

OptionSymbolFromDescriptionRequest

Field Name	Data Type	Accepted Values/Examples
Expiration	string	Time at which order is no longer valid (DAY, DAYPLUS, or custom value supported by destination)
OptionType	OptionTypes	Refer to the section OptionTypes
Root	string	Symbol root for option
StrikePrice	double	Price at which option can be exercised
UserToken	string	A server generated GUID that is given as response to the client during the first login

OptionSymbolFromDescriptionResponse

Field Name	Data Type	Retrieves
Acknowledgement	ServerAcknowledgement	Response from the server (success or failure). Refer to the section ServerAcknowledgement

Field Name	Data Type	Retrieves
Symbol	string	Valid ticker symbol (e.g., AAPL, IBM, VOD.LSE)

GetSecurityData

The **SecurityDataRequest** object is used as an input in **GetSecurityData** API. This API is used to request security data.

SecurityDataRequest

Field Name	Data Type	Accepted Values/Examples
Symbol	string	Valid ticker symbol (e.g., AAPL, IBM, VOD.LSE)
Symbols	repeated string	Valid ticker symbols (e.g., AAPL, IBM, VOD.LSE). This field takes precedence over Symbol field.
UserToken	string	A server generated GUID that is given as response to the client during the first login

SecurityDataResponse

Field Name	Data Type	Retrieves
Acknowledgement	ServerAcknowledgement	Response from the server (success or failure). Refer to the section ServerAcknowledgement
SecurityInfoList	repeated SecurityData	Refer to the section SecurityData

GetSymbolFromAlternateSymbology

The **SymbolFromAlternateSymbologyRequest** object is used as an input in **GetSymbolFromAlternateSymbology** API. This API is used to fetch Symbols using SEDOL, CUSIP or ISIN code.

SymbolFromAlternateSymbologyRequest

Field Name	Data Type	Accepted Values/Examples
Symbol	string	Valid ticker symbol (e.g., AAPL, IBM, VOD.LSE)
SymbolInfo	AlternateSymbology	Refer to the section AlternateSymbology

Field Name	Data Type	Accepted Values/Examples
Symbols	repeated string	Valid ticker symbols (e.g., AAPL, IBM, VOD.LSE). This field takes precedence over Symbol field.
UserToken	string	A server generated GUID that is given as response to the client during the first login

SymbolFromAlternateSymbologyResponse

Field Name	Data Type	Retrieves
Acknowledgement	ServerAcknowledgement	Response from the server (success or failure). Refer to the section ServerAcknowledgement
FailureCount	int32	Count of failed lookups (bulk only)
Results	repeated SymbolLookupResult	Fetch bulk symbol results
SuccessCount	int32	Count of successful lookups (bulk only)
SymbolInfolist	repeated SymbolData	Refer to the section SymbolData
TotalCount	int32	Total count of bulk symbols requested

GetSymbolReferenceData

The **SymbolReferenceDataRequest** object is used as an input in **GetSymbolReferenceData** API. This API is used to request symbol reference data.

SymbolReferenceDataRequest

Field Name	Data Type	Accepted Values/Examples
Symbol	string	Valid ticker symbol (e.g., AAPL, IBM, VOD.LSE)
UserToken	string	A server generated GUID that is given as response to the client during the first login

SymbolReferenceDataResponse

Field Name	Data Type	Retrieves
Acknowledgement	ServerAcknowledgement	Response from the server (success or failure). Refer to the section ServerAcknowledgement
SymInfoList	repeated SymInfoResponse	Refer to the section SymInfoResponse

GetSymbolsFromCompanyName

The **SymbolsFromCompanyNameRequest** object is used as an input in **GetSymbolsFromCompanyName** API. This API is used to fetch symbols for a company you entered.

SymbolsFromCompanyNameRequest

Field Name	Data Type	Accepted Values/Examples
CompanyName	string	Company name represented by symbol
UserToken	string	A server generated GUID that is given as response to the client during the first login

SymbolsFromCompanyNameResponse

Field Name	Data Type	Retrieves
Acknowledgement	ServerAcknowledgement	Response from the server (success or failure). Refer to the section ServerAcknowledgement
SymbolDatalist	repeated SymbolData	Refer to the section SymbolData

GetTickData

The **TickDataRequest** object is used as an input in **GetTickData** API. This API is used to request tick data.

TickDataRequest

Field Name	Data Type	Accepted Values/Examples
Date	google.protobuf.Timestamp	Trade date
RequestId	google.protobuf.Int32Value	A user-supplied request ID that can be used to dis-ambiguate response data
StartTime	google.protobuf.Duration	The time trading started for the current session
StopTime	google.protobuf.Duration	The time trading ended
Symbol	string	Valid ticker symbol (e.g., AAPL, IBM, VOD.LSE)
TickTypes	repeated TickTypes	Refer to the section TickTypes
UserToken	string	A server generated GUID that is given as response to the client during the first login

TickDataResponse

Field Name	Data Type	Retrieves
Acknowledgement	ServerAcknowledgement	Response from the server (success or failure). Refer to the section ServerAcknowledgement
TickInfo	repeated TicksResponse	Refer to the section TicksResponse

RemoveSymbols

The **RemoveSymbolsRequest** object is used as an input in **RemoveSymbols** API. This API is used to remove symbols from your account.

RemoveSymbolsRequest

Field Name	Data Type	Accepted Values/Examples
MarketDataLevel	string	Level1 or Level2
Symbols	repeated string	Valid ticker symbols
UserToken	string	A server generated GUID that is given as response to the client during the first login

RemoveSymbolsResponse

Field Name	Data Type	Retrieves
Acknowledgement	ServerAcknowledgement	Response from the server (success or failure). Refer to the section ServerAcknowledgement
OptionalFields	map<string, string>	This is additional information given to the client in cases of failure. The field OptionalFields ["ErrorMessage"] would contain the reason for error or exception
ServerResponse	string	Success: In case the invoked API executes successfully on the server side Failed: In case the invoked API throws an error or lands in an exception

StreamMarketData

The **MarketDataStreamRequest** object is used as an input to the **StreamMarketData** API. This API provides streaming access to market data, including Level 1, Level 2, or Tick data for a given symbol. You

can also add additional symbols to the stream without needing to unsubscribe from the currently subscribed symbols.

MarketDataStreamRequest

Field Name	Data Type	Accepted Values/Examples
Advise	bool	If set to **True**, real-time updates from the server will be registered for
MarketDataLevel	string	LEVEL1, LEVEL2, TICK, INTRADAY_BARS
MarketSource	repeated string	A list of market sources to which the returned data should be restricted. If empty, all sources are included
RegionalExchangeIds	repeated string	Regional exchange ID
Request	bool	If set to **True**, a current snapshot of the data will be retrieved
RequestType	string	To add, remove, or change a symbol(s) subscription
Symbols	string	Valid ticker symbol (e.g., AAPL, IBM, VOD.LSE)
UserToken	string	A server generated GUID that is given as response to the client during the first login

MarketDataStreamResponse

Field Name	Data Type	Retrieves
Acknowledgement	ServerAcknowledgement	Response from the server (success or failure). Refer to the section ServerAcknowledgement
Ask	Price	Last ask price for symbol
Bid	Price	Last bid price for symbol
ChangeLast	Price	Change from last price
CompanyName	string	Company name represented by symbol
DispName	string	'IBM' (Symbol formats are datafeed-specific. Check your feed handler help file for the correct format.)
ExchName	string	Three-letter acronym for listing exchange

Field Name	Data Type	Retrieves
High1	Price	Value of highest transaction for current session
Low1	Price	Value of lowest transaction for current session
MktMkrAsksize	google.protobuf.Int32Value	Market maker ask size (LME)
MktMkrBidsize	google.protobuf.Int32Value	Market maker bid size (LME)
MktMkrDisplay	string	Market maker display
MktMkrId	string	Market maker ID
MktMkrStatus	string	Market maker status
MktSource	string	Market maker quote source
ResponseType	string	LEVEL1_DATA, LEVEL2_DATA, TICK_DATA, INTRADAY_BAR_DATA, STATUS_UPDATE
SymbolDesc	string	Symbol description
Trdprc1	Price	Refer to the section TicksResponse
TrdTim1	repeated google.protobuf.Duration	Time of last trade
MktMkrBid	Price	Market maker bid price (LME)
MktMkrAsk	Price	Market maker ask price (LME)
TrdPrc1List	repeated Price	List of last traded price
TrdDateList	repeated google.protobuf.Timestamp	List of date of last trade
TickTypeList	repeated int32	List of tick type
TrdXid1List	repeated string	List of exchange ID of last trade
TrdVol1List	repeated int32	List of the volume of last reported trade
TrdTim1List	repeated google.protobuf.Duration	List of time of last trade
Count	int32	Tick count

Field Name	Data Type	Retrieves
StatusMessage	string	Status of the symbols subscribed to in the market data stream

SubscribeIntradayBars

The **SubscribeIntradayBarsRequest** object is used as an input in **SubscribeIntradayBars** API. This API is used to subscribe to the intraday bars.

SubscribeIntradayBarsRequest

Field Name	Data Type	Accepted Values/Examples
BarInterval	int32	Bar interval in minutes
DaysBack	google.protobuf.Int32Value	The number of days worth of bars to retrieve
EmitPartialBars	bool	If set to True, partial (incomplete) intra-day bars are emitted as they are being formed, before the bar interval is com-plete. If set to False, only fully com-pleted bars are emitted
PartialIntervalSeconds	google.protobuf.Int32Value	Specifies the interval in seconds at which partial intraday bar updates are emitted. This field is applicable only when EmitPartialBars is set to True.
RequestId	google.protobuf.Int32Value	A user-supplied request ID that can be used to disambiguate response data
ShowPrePostMarket	bool	If set to True, displays the pre/post trad-ing hours market data
SinceTimestamp	google.protobuf.Timestamp	Since the time stamp
StartAtMidnight	bool	If set to True, the first bar of the day includes all data since midnight, oth-erwise the first bar of the day includes all data since the session open time
Symbols	repeated string	Valid ticker symbols
UserToken	string	A server generated GUID that is given as response to the client during the first login

SubscribeIntradayBarsResponse

Field Name	Data Type	Retrieves
Acknowledgement	ServerAcknowledgement	Response from the server (success or failure). Refer to the section ServerAcknowledgement
BarInterval	int32	Bar interval in minutes
Bars	repeated IntradayBarUpdate	Retrieves the intraday bar data for the subscribed symbol. Refer to the section CommonBarsFields
DispName	string	'IBM' (Symbol formats are datafeed-specific. Check your feed handler help file for the correct format.)

SubscribeLevel1Ticks

The **Level1MarketDataRequest** object is used as an input in **SubscribeLevel1Ticks** API. This API is used to fetch the level 1 market data snapshot.

Level1MarketDataRequest

Field Name	Data Type	Accepted Values/Examples
Advise	bool	If set to **True**, real-time updates from the server will be registered for
RegionalExchangeIds	repeated string	Regional exchange ID
Request	bool	If set to **True**, a current snapshot of the data will be retrieved
Symbols	repeated string	Valid ticker symbol (e.g., AAPL, IBM, VOD.LSE)
UserToken	string	A server generated GUID that is given as response to the client during the first login

Level1MarketDataResponse

Field Name	Data Type	Retrieves
Acknowledgement	ServerAcknowledgement	Response from the server (success or failure). Refer to the section ServerAcknowledgement

Field Name	Data Type	Retrieves
ArcaImbalanceVolume	google.protobuf.Int32Value	Pre-Open imbalance volume (positive is bid, negative is offer)
ArcaMatchVolume	google.protobuf.Int32Value	Total number of shares to be executed
Ask	Price	Last ask price for symbol
Bid	Price	Last bid price for symbol
BidTick2	string	"Bid tick" for NASDAQ short sale rule
ChangeLast	Price	Change from last price
CompanyName	string	Company name represented by symbol
DispName	string	'IBM' (Symbol formats are datafeed-specific. Check your Feed Handler help file for the correct format.)
High1	Price	Value of highest transaction for current session
High52	Price	Highest traded price for a share in the last 52 weeks
IntradayHighCount	google.protobuf.Int32Value	Number of intraday highs
Low1	Price	Value of lowest transaction for current session
Low52	Price	Lowest traded price for a share in the last 52 weeks
LrpAsk	Price	NYSE liquidity replenishment point for Asks
LrpBid	Price	NYSE liquidity replenishment point for bids
Open Interest	Long	Open interest
SaleConditionVolume	google.protobuf.Int32Value	Volume of the most recent pre-market, post-market, or irregular trade
SymbolDesc	string	Symbol description
Trdprc1	Price	Last traded price

Field Name	Data Type	Retrieves
Trdtim1	google.protobuf.Duration	Trade time
Vwap	google.protobuf.DoubleValue	VWAP from the trade server
VwapVol	google.protobuf.Int32Value	VWAP Volume from the trade server

SubscribeLevel2Ticks

The **Level2MarketDataRequest** object is used as an input in **SubscribeLevel2Ticks** API. This API is used to acquire the level 2 market data information.

Level2MarketDataRequest

Field Name	Data Type	Accepted Values/Examples
Advise	bool	If set to **True**, real-time updates from the server will be registered for
MarketSource	repeated string	A list of market sources to which the returned data should be restricted. If empty, all sources are included
Request	bool	If set to **True**, a current snapshot of the data will be retrieved
RequestId	google.protobuf.Int32Value	A user-supplied request ID that can be used to dis-ambiguate response data
Symbols	repeated string	Valid ticker symbol (e.g., AAPL, IBM, VOD.LSE)
UserToken	string	A server generated GUID that is given as response to the client during the first login

Level2MarketDataResponse

Field Name	Data Type	Retrieves
Acknowledgement	ServerAcknowledgement	Response from the server (success or fail-ure). Refer to the section Server-Acknowledgement
DispName	string	'IBM' (Symbol formats are datafeed-spe-cific. Check your feed handler help file for the correct format.)

Field Name	Data Type	Retrieves
ExchName	string	Three-letter acronym for listing exchange
MktMkrAsk	Price	Market maker ask price (LME)
MktMkrAsksize	google.protobuf.Int32Value	Market maker ask size (LME)
MktMkrAskTime	google.protobuf.Duration	Market maker ask time (LME)
MktMkrBid	Price	Market maker bid price (LME)
MktMkrBidsize	google.protobuf.Int32Value	Market maker bid size (LME)
MktMkrBidTime	google.protobuf.Duration	Market maker bid time (LME)
MktMkrChangeLastAsk	Price	Ask change
MktMkrDate	google.protobuf.Timestamp	Market maker date (LME)
MktMkrDisplay	string	Market maker display
MktMkrId	string	Market maker ID
MktMkrStatus	string	Market maker status
MktSource	string	Market maker quote source
QuoteUpdateType	google.protobuf.Int32Value	Type of update (live, refresh, or recovery)
Styp	google.protobuf.Int32Value	Security type
SymbolError	google.protobuf.Int32Value	Symbol error
SymbolId	google.protobuf.Int32Value	Symbol ID
TableId	google.protobuf.Int32Value	Table ID

SubscribeTickData

The **SubscribeTickDataRequest** object is used as an input in **SubscribeTickData** API. This API is used to fetch a streaming tick data.

SubscribeTickDataRequest

Field Name	Data Type	Accepted Values/Examples
Date	google.protobuf.Timestamp	Trade date

Field Name	Data Type	Accepted Values/Examples
Request	bool	If set to **True**, a current snapshot of the data will be retrieved
RequestId	google.protobuf.Int32Value	A user-supplied request ID that can be used to disambiguate response data
StartTime	google.protobuf.Duration	The time trading started for the current session
StopTime	google.protobuf.Duration	The time trading ended
Symbol	string	Valid ticker symbol (e.g., AAPL, IBM, VOD.LSE)
TickTypes	repeated TickTypes	Refer to the section TickTypes
UserToken	string	A server generated GUID that is given as response to the client during the first login

SubscribeTickDataResponse

Field Name	Data Type	Retrieves
Acknowledgement	ServerAcknowledgement	Response from the server (success or failure). Refer to the section ServerAcknowledgement
Count	int32	Tick count
DispName	repeated string	'IBM' (Symbol formats are datafeed-specific. Check your feed handler help file for the correct format.)
TickInfo	repeated TicksResponse	Refer to the section TicksResponse
TickType	repeated int32	Tick type
TrdDate	repeated google.protobuf.Timestamp	Date of last trade
TrdTim1	repeated google.protobuf.Duration	Time of last trade
TrdVol1	repeated int32	Volume of last reported trade
TrdXid1	repeated string	Exchange ID of last trade

UnSubscribeIntradayBars

The **UnSubscribeIntradayBarsRequest** object is used as an input in **UnSubscribeIntradayBars** API. This API allows you to unsubscribe from intraday bars for a specific symbol.

UnSubscribeIntradayBarsRequest

Field Name	Data Type	Accepted Values/Examples
UserToken	string	A server generated GUID that is given as response to the client during the first login

UnSubscribeIntradayBarsResponse

Field Name	Data Type	Retrieves
Acknowledgement	ServerAcknowledgement	Response from the server (success or failure). Refer to the section ServerAcknowledgement

UnSubscribeLevel1Data

The **UnSubscribeLevel1DataRequest** object is used as an input in **UnSubscribeLevel1Data** API. This API is used to unsubscribe the level 1 data.

UnSubscribeLevel1DataRequest

Field Name	Data Type	Accepted Values/Examples
UserToken	string	A server generated GUID that is given as response to the client during the first login

UnSubscribeLevel1DataResponse

Field Name	Data Type	Retrieves
Acknowledgement	ServerAcknowledgement	Response from the server (success or failure). Refer to the section ServerAcknowledgement
OptionalFields	map<string, string>	This is additional information given to the client in cases of failure. The field OptionalFields [“ErrorMessage”] would contain the reason for error or exception
ServerResponse	string	Success: In case the invoked API executes successfully on the server side Failed: In case the invoked API throws an error or lands in an exception

UnSubscribeLevel2Data

The **UnSubscribeLevel2DataRequest** object is used as an input in **UnSubscribeLevel2Data** API. This API is used to unsubscribe the level 2 data.

UnSubscribeLevel2DataRequest

Field Name	Data Type	Accepted Values/Examples
UserToken	string	A server generated GUID that is given as response to the client during the first login

UnSubscribeLevel2DataResponse

Field Name	Data Type	Retrieves
Acknowledgement	ServerAcknowledgement	Response from the server (success or failure). Refer to the section ServerAcknowledgement
OptionalFields	map<string, string>	This is additional information given to the client in cases of failure. The field OptionalFields [“ErrorMessage”] would contain the reason for error or exception
ServerResponse	string	Success: In case the invoked API executes successfully on the server side Failed: In case the invoked API throws an error or lands in an exception

UnSubscribeStreamMarketData

The **UnSubscribeStreamMarketDataRequest** object is used as an input to the **UnSubscribeStreamMarketData** API. This API allows you to unsubscribe from market data for a specific symbol, including Level 1, Level 2, or Tick data. You can dynamically remove one or more symbols from the active data stream.

UnSubscribeStreamMarketDataRequest

Field Name	Data Type	Accepted Values/Examples
UserToken	string	A server generated GUID that is given as response to the client during the first login

UnSubscribeStreamMarketDataResponse

Field Name	Data Type	Retrieves
Acknowledgement	ServerAcknowledgement	Response from the server (success or failure). Refer to the section ServerAcknowledgement
OptionalFields	map<string, string>	This is additional information given to the client in cases of failure. The field OptionalFields ["ErrorMessage"] would contain the reason for error or exception
ServerResponse	string	Success: In case the invoked API executes successfully on the server side Failed: In case the invoked API throws an error or lands in an exception

UnSubscribeTickData

The **UnSubscribeTickDataRequest** object is used as an input in **UnSubscribeTickData** API. This API is used to unsubscribe the tick data.

UnSubscribeTickDataRequest

Field Name	Data Type	Accepted Values/Examples
UserToken	string	A server generated GUID that is given as response to the client during the first login

UnSubscribeTickDataResponse

Field Name	Data Type	Retrieves
Acknowledgement	ServerAcknowledgement	Response from the server (success or failure). Refer to the section ServerAcknowledgement

Order APIs

The following APIs are provided as part of Order:

API Name	Input	Output
Single Order APIs		
CancelSingleOrder	CancelSingleOrderRequest	CancelSingleOrderResponse
ChangeSingleOrder	ChangeSingleOrderRequest	ChangeSingleOrderResponse
SubmitAllocationOrder	SubmitAllocationOrderRequest	SubmitAllocationOrderResponse
SubmitBookTrade	SubmitBookTradeRequest	SubmitBookTradeResponse
SubmitSingleOrder	SubmitSingleOrderRequest	SubmitSingleOrderResponse
SubmitTradeReport	SubmitTradeReportRequest	SubmitTradeReportResponse
Pair Order APIs		
CancelPairOrder	CancelPairOrderRequest	CancelPairOrderResponse
ChangePairOrder	ChangePairOrderRequest	ChangePairOrderResponse
SubmitPairOrder	SubmitPairOrderRequest	SubmitPairOrderResponse
Basket Order APIs		
SubmitBasketOrder	BasketOrderRequest	BasketOrderResponse
Miscellaneous APIs		
GetOrderDetailByDateRange	OrderDetailByDateRangeRequest	stream Order-DetailByDateRangeResponse
GetOrderDetailByOrderId	OrderDetailByOrderIdRequest	OrderDetailByOrderIdResponse
GetOrderDetailByOrderTag	OrderDetailByOrderTagRequest	OrderDetailByOrderTagResponse
GetUserAccounts	UserAccountsRequest	UserAccountsResponse
SubmitSeedData	SubmitSeedDataRequest	SubmitSeedDataResponse
SubscribeOrderInfo	SubscribeOrderInfoRequest	stream Sub-scribeOrderInfoResponse
SubscribeOrderInfoJson	SubscribeOrderInfoJsonRequest	stream Sub-scribeOrderInfoJsonResponse

Custom Data Type

AllocationDetails

Field Name	Data Type	Accepted Values/Examples
AllocationDestinationRoute	string	Destination route for the allocated order
CommissionRate	google.protobuf.DoubleValue	Commission rate
CommissionRateType	google.protobuf.Int32Value	Commission rate type
GeneralMessage	string	User provided general text
NetPrice	google.protobuf.DoubleValue	Net price
TargetAccount	string	Destination account for the allocated order
TargetPrice	google.protobuf.DoubleValue	Limit price of allocated order
TargetQuantity	int32	Volume of the ticket if the ticket is live, or the Traded Amount if the order has been canceled before completion

AllocationType

Field Name	Data Type	Accepted Values/Examples
Allocation	enum	UserSubmitAllocation = 0; UserSubmitAllocationEx = 1

OrderDetailsResponse

Field Name	Data Type	Accepted Values/Examples
Account	string	Semi-colon separated values that represent either Trade or Neutral accounts the user has permission to e.g., TAL;TEST;USER1;TRADE or TAL;TEST;USER2;NEUTRAL
Buyorsell	string	BUY, SELL, or SELLSHORT. This field will soon be deprecated. Alternatively, you can use the field Side

Field Name	Data Type	Accepted Values/Examples
ClaimedByClerk	string	Claimed by clerk
CurrentStatus	string	Current status of order (PENDING, LIVE, COMPLETED, or DELETED)
ExpirationDate	google.protobuf.Timestamp	Date at which order is no longer valid
GoodFrom	google.protobuf.Timestamp	Time at which order is first valid for execution
LinkedOrderId	string	ID of linked order
OrderId	string	A unique id associated to every order. This is the identifier to lookup the specific order to modify
OrderTag	string	Order tag
OriginalOrderId	string	Original order identification code
PairSpreadType	string	Pair spread type
Price	google.protobuf.DoubleValue	Limit price submitted in order
PriceType	PriceTypeEnum	By default the price type is set to Market, in case no value is provided. Refer to the section PriceTypeEnum
Reason	string	Reason given by user or destination for chan-ging, can-celling, or deleting the order
RefersToId	string	Refers to the identification code
Route	string	Route name as shown in Eze EMS. Note: This field is also referred to as **Exit Vehicle**

Field Name	Data Type	Accepted Values/Examples
Side	string	BUY, SELL, or SELLSHORT. To send a sell short order, assign the SHORT_LOCATE_ID extended field. To send a buy to cover order, set the side to BUY and assign the TO_OPEN_POS extended field: • 100: Open a new position • 101: Close a position • When omitted, the system automatically determines position intent based on Order side (buy or sell), User account trading permissions, Existing positions
StopPrice	google.protobuf.DoubleValue	Stop price
submitTime	google.protobuf.Timestamp	The time when the trade was submitted
Symbol	string	Valid ticker symbol (e.g., AAPL, IBM, VOD.LSE)
TicketId	string	Ticket ID
TimeInForce	ExpirationType	Time or date at which order is no longer valid. In case no value is provided, **DAY** expiration type is set by default. Refer to the section ExpirationType
TimeStamp	string	Timestamp applied to order by trading system
TraderId	string	USERNAME@DOMAIN of the message recipient
Type	string	Order Event Type. Refer to the section Appendix A
UserMessage	string	User message/notes
Volume	int32	Order quantity
VolumeTraded	int32	Volume traded so far by the order

OrderInfoFilters

Field Name	Data Type	Accepted Values/Examples
IncludeClerkReject	bool	Boolean flag to filter the order responses based on order type ClerkReject
IncludeExchangeKillOrder	bool	Boolean flag to filter the order responses based on order type ExchangeKillOrder
IncludeExecutions	bool	If set to **True** you get ExchangeAcceptOrder, ExchangeTradeOrder, ExchangeBustOrder events in a DataFrame with row by row format as response If you wish to learn more about these order event types refer to the section Appendix A: Order Event Types
IncludeOrders	bool	If set to **True** you get UserSubmitOrder, User-SubmitStagedOrder, UserSubmitCompoundOrder, User-SubmitChange, UserSubmitCancel, UserSubmitAllocation, UserSubmitAllocationEx, For-eignExecution order events in a DataFrame with row by row format as response If you wish to learn more about these order event types refer to the section Appendix A: Order Event Types

OrderRow

Field Name	Data Type	Accepted Values/Examples
AcctType	google.protobuf.Int32Value	Type of account for a position
AllocatedValue	google.protobuf.DoubleValue	Allocated value
Ask	string	Last ask price for symbol
AvgPrice	google.protobuf.DoubleValue	Average execution price for order so far
Bank	string	Bank of trading account for selected position
Basisvalue	google.protobuf.DoubleValue	Currency value of a unit move
Bid	string	Last bid price for symbol
BookId	google.protobuf.Int32Value	Book ID

Field Name	Data Type	Accepted Values/Examples
Branch	string	Branch of trading account for selected position
Buyorsell	string	BUY, SELL, or SELLSHORT
ClaimedByClerk	string	Claimed by clerk
ClientOrderId	string	Client order ID
Commission	google.protobuf.DoubleValue	Net of commissions for today's trades for a symbol
CommissionCode	string	Broker code
CommissionRate	google.protobuf.DoubleValue	Commission rate
CommissionRateType	google.protobuf.Int32Value	Commission rate type
CrossFlag	google.protobuf.Int32Value	Cross flag
Currency	string	Currency of symbol being traded or Currency of position
CurrentStatus	string	Current status of order (PENDING, LIVE, COMPLETED, or DELETED)
Customer	string	Customer name of trading account for selected position
DateIndex	google.protobuf.Int32Value	Date index
Deposit	string	Deposit or account name for selected position
DispName	string	'IBM' (Symbol formats are datafeed-specific. Check your feed handler help file for the correct format.)
EcnFee	google.protobuf.DoubleValue	ECN fee
Exchange	string	Three-letter acronym for listing exchange
ExecutionState	google.protobuf.Int32Value	Execution state
ExpirDate	google.protobuf.Timestamp	Date of expiration for an option

Field Name	Data Type	Accepted Values/Examples
ExtendedStateFlags	google.protobuf.Int32Value	Extended state information for internal use
ExtendedStateFlags2	google.protobuf.Int32Value	Extended state flags 2
ExternalAcceptanceFlag	google.protobuf.Int32Value	External acceptance flag
FixTraderId	string	FIX trader ID
FornexSourceFlags	google.protobuf.Int32Value	Forex source
GoodFrom	string	Time at which order is first valid for execution
GoodUntil	string	Time at which order is no longer valid (DAY, DAYPLUS, or custom value supported by destination)
GwBookSeqNo	google.protobuf.Int32Value	Gateway book sequence number
Latency3	google.protobuf.DoubleValue	Latency
Latency6	google.protobuf.DoubleValue	Latency
LinkedOrderCancellation	google.protobuf.Int32Value	Linked order cancellation
LinkedOrderId	string	ID of linked order
LinkedOrderRelationship	google.protobuf.Int32Value	Linked order relationship
Minmove	google.protobuf.Int32Value	Minimum movement
NewRemoteId	string	New remote identification code
NewsDate	google.protobuf.Timestamp	Date of most recent news story
NewsTime	google.protobuf.Duration	Time of most recent news story
OmsClientType	google.protobuf.Int32Value	OMS client type
OppositeParty	string	Contra (used by ARCAEX to signify liquidity added or removed)
OrderFlags	google.protobuf.Int32Value	Order flags
OrderFlags2	google.protobuf.Int32Value	Order flag 2

Field Name	Data Type	Accepted Values/Examples
OrderId	string	A unique id associated to every order. This is the identifier to lookup the specific order to cancel
OrderResidual	google.protobuf.Int32Value	Residual volume
OrderTag	string	Order tag
OriginalOrderId	string	Original order identification code
OriginalPrice	string	Original price
OriginalTraderId	string	USERNAME@DOMAIN of the trader who placed the trade originally
OriginalVolume	google.protobuf.Int32Value	Original volume of trade
PairCash	google.protobuf.DoubleValue	Pair cash component
PairImbalanceLimit	google.protobuf.DoubleValue	Pair imbalance limit
PairImbalanceLimitType	google.protobuf.Int32Value	Pair imbalance limit type
PairLeg1Benchmark	google.protobuf.DoubleValue	Pair Leg 1 benchmark
PairLeg1BenchmarkType	google.protobuf.Int32Value	Pair Leg 1 benchmark
PairLeg2Benchmark	google.protobuf.DoubleValue	Pair Leg 2 benchmark
PairLeg2BenchmarkType	google.protobuf.Int32Value	Pair Leg 2 benchmark
PairRatio	google.protobuf.DoubleValue	Pair ratio
PairSpread	google.protobuf.DoubleValue	Pair spread
PairSpreadType	google.protobuf.Int32Value	Pair spread type
PairTarget	google.protobuf.DoubleValue	Pair target
Price	string	Limit price submitted in order
PriceType	string	By default the price type is set to **Market**, in case no value is provided. Refer to the section PriceTypeEnum
Putcallind	string	Option type (P = put, C = call, U = underlier) for symbol

Field Name	Data Type	Accepted Values/Examples
Rank	google.protobuf.Int32Value	Rank
Reason	string	Reason given by user or destination for changing, cancelling, or deleting the order
RefersToId	string	Refers to the identification code
RemainingVolume	google.protobuf.Int32Value	Remaining volume
RemoteId	string	Remote ID
Route	string	Route name as shown in Eze EMS. Note: This field is also referred to as **Exit Vehicle**
SharesAllocated	google.protobuf.Int32Value	Shares allocated
ShortLocateId	string	ID assigned to short sell orders as required by regulation SHO
SpreadClip	google.protobuf.DoubleValue	Spread clip
SpreadClipType	google.protobuf.Int32Value	Spread clip type
SpreadLegCount	google.protobuf.Int32Value	Number of legs this spread contains
SpreadLegLeanPriority	google.protobuf.Int32Value	Spread leg lean priority
SpreadLegNumber	google.protobuf.Int32Value	Number of legs this spread contains
SpreadLegPriceType	google.protobuf.Int32Value	Spread leg price type
SpreadNumLegs	google.protobuf.Int32Value	Spread number of legs
StopPrice	string	Stop price
StrikePrc	string	Price at which option can be exercised
Styp	google.protobuf.Int32Value	Security type
Table	string	Table
TboAccountId	google.protobuf.Int32Value	Account ID
TicketId	string	Ticket ID

Field Name	Data Type	Accepted Values/Examples
TimeStamp	string	Timestamp applied to order by trading system
TraderCapacity	string	Trader capacity
TraderId	string	USERNAME@DOMAIN of the message recipient
TrdTime	google.protobuf.Duration	Time of last trade
Type	string	Order Event Type. Refer to the section Appendix A
Undersym	string	Underlier symbol
UserMessage	string	User message/notes
UtcOffset	google.protobuf.Int32Value	UTC offset
Volume	google.protobuf.Int32Value	Order quantity
VolumeTraded	google.protobuf.Int32Value	Volume traded so far by the order
VolumeType	string	Indicates the fill type of the position for the symbol
WorkingQty	google.protobuf.Int32Value	Quantity out in the market

OrderType

Field Name	Data Type	Accepted Values/Examples
Type	enum	UserSubmitOrder = 0; ForeignExecution = 1

SubscribeOrderFieldOptions

Field Name	Data Type	Accepted Values/Examples
Fields	enum	SUBMIT_TIME = 0 ORDER_ID = 1 LINKED_ORDER_ID = 2 REFERS_TO_ID = 3 TICKET_ID = 4 ORIGINAL_ORDER_ID = 5 SYMBOL = 6 TYPE = 7 CURRENT_STATUS = 8 TRADER_ID = 9 CLAIMED_BY_CLERK = 10 VOLUME = 11 PRICE = 12 PRICE_TYPE = 13 BUYORSELL = 14 PAIR_SPREAD_TYPE = 15 REASON = 16 TIME_STAMP = 17 EXTENDED_FIELDS = 18 GOOD_FROM = 19 TIME_IN_FORCE = 20 STOP_PRICE = 21 USER_MESSAGE = 22 EXPIRATION_DATE = 23 SIDE = 24 ROUTE = 25 ACCOUNT = 26 ORDER_TAG = 27 VOLUME_TRADED = 28
RequestedExtendedFields	repeated string	Fetch details for the specified extended fields
RequestedFields	repeated Fields	Fetch details for the specified fields

TradeType

Field Name	Data Type	Accepted Values/Examples
TypeofOrder	enum	UserSubmitTradeReport = 0; ForeignExecution = 1

CancelPairOrder

The **CancelPairOrderRequest** object is used as an input in **CancelPairOrder** API. This API is used to cancel a pair order using its order ID.

CancelPairOrderRequest

Field Name	Required?	Data Type	Accepted Values/Examples
OrderId	Yes	string	A unique id associated to every order. This is the identifier to lookup the specific order to cancel. Note: In case of **CancelPairOrder** the value of OrderId will be that of the **Pair** and not that of the individual orders
UserToken	Yes	string	A server generated GUID that is given as response to the client during the first login

CancelPairOrderResponse

Field Name	Data Type	Retrieves
Acknowledgement	ServerAcknowledgement	Response from the server (success or failure). Refer to the section ServerAcknowledgement
OptionalFields	map<string, string>	This is additional information given to the client in cases of failure. The field OptionalFields ["ErrorMessage"] would contain the reason for error or exception
ServerResponse	string	Success: In case the invoked API executes successfully on the server side Failed: In case the invoked API throws an error or lands in an exception

CancelSingleOrder

The **CancelSingleOrderRequest** object is used as an input in **CancelSingleOrder** API. This API is used to cancel a single order using its order ID.

CancelSingleOrderRequest

Field Name	Required?	Data Type	Accepted Values/Examples
OrderId	Yes	string	A unique id associated to every order. This is the identifier to lookup the specific order to cancel

Field Name	Required?	Data Type	Accepted Values/Examples
UserToken	Yes	string	A server generated GUID that is given as response to the client during the first login

CancelSingleOrderResponse

Field Name	Data Type	Retrieves
Acknowledgement	ServerAcknowledgement	Response from the server (success or failure). Refer to the section ServerAcknowledgement
OptionalFields	map<string, string>	This is additional information given to the client in cases of failure. The field OptionalFields ["ErrorMessage"] would contain the reason for error or exception
ServerResponse	string	Success: In case the invoked API executes successfully on the server side Failed: In case the invoked API throws an error or lands in an exception

ChangePairOrder

The **ChangePairOrderRequest** object is used as an input in **ChangePairOrder** API. This API can be used to modify a broker specific pair algo order using its order ID.

ChangePairOrderRequest

Field Name	Required?	Data Type	Accepted Values/Examples
Leg1_Quantity	No	int32	Value > 0
Leg1OrderId	Yes	string	A unique id associated to every order. This is the identifier to lookup the specific Leg1 order to modify
Leg2_Quantity	No	int32	Value > 0
Leg2OrderId	Yes	string	A unique id associated to every order. This is the identifier to lookup the specific Leg2 order to modify
PairOrderId	Yes	string	A unique id associated to every order. This is the identifier to lookup the specific pair order to modify
UserToken	Yes	string	A server generated GUID that is given as response to the client during the first login

ChangePairOrderResponse

Field Name	Data Type	Retrieves
Acknowledgement	ServerAcknowledgement	Response from the server (success or failure). Refer to the section ServerAcknowledgement
OptionalFields	map<string, string>	This is additional information given to the client in cases of failure. The field OptionalFields ["ErrorMessage"] would contain the reason for error or exception
ServerResponse	string	Success: In case the invoked API executes successfully on the server side Failed: In case the invoked API throws an error or lands in an exception

ChangeSingleOrder

The **ChangeSingleOrderRequest** object is used as an input in **ChangeSingleOrder** API. This API can be used to modify the parameters (volume, price type, expiration type etc.) of a single order using its order ID.

ChangeSingleOrderRequest

Field Name	Required?	Data Type	Accepted Values/Examples
ExpirationDate	No	google.protobuf.Timestamp	Date at which order is no longer valid
GoodFrom	No	google.protobuf.Timestamp	Time at which order is first valid for execution
OrderId	Yes	string	A unique id associated to every order. This is the identifier to lookup the specific order to modify
OrderTag	No	string	Order tag
Price	No	google.protobuf.DoubleValue	Limit price submitted in order
PriceType	No	PriceTypeEnum	By default the price type is set to **Market**, in case no value is provided. Refer to the section PriceTypeEnum
Quantity	No	int32	Value > 0

Field Name	Required?	Data Type	Accepted Values/Examples
StopPrice	No	google.protobuf.DoubleValue	Stop price
TimeInForce	No	ExpirationType	Time or date at which order is no longer valid. In case no value is provided, **DAY** expiration type is set by default. Refer to the section ExpirationType
UserMessage	No	string	User message/notes
UserToken	Yes	string	A server generated GUID that is given as response to the client during the first login

ChangeSingleOrderResponse

Field Name	Data Type	Retrieves
Acknowledgement	ServerAcknowledgement	Response from the server (success or failure). Refer to the section ServerAcknowledgement
OptionalFields	map<string, string>	This is additional information given to the client in cases of failure. The field OptionalFields ["ErrorMessage"] would contain the reason for error or exception
ServerResponse	string	Success: In case the invoked API executes successfully on the server side Failed: In case the invoked API throws an error or lands in an exception

GetOrderDetailByDateRange

The **OrderDetailByDateRangeRequest** object is used as an input in **GetOrderDetailByDateRange** API. This API is used to fetch list of orders placed between the given date range. The maximum date range that can be provided in a single request is 1 year. Order detail history is maintained up to three years.

OrderDetailByDateRangeRequest

Field Name	Required?	Data Type	Accepted Values/Examples
Account	No	string	Semi-colon separated values that represent either Trade or Neutral accounts the user has permission to e.g., TAL;TEST;USER1;TRADE or TAL;TEST;USER2;NEUTRAL Note: Use this field with **Start Date** and **End Date** to retrieve order details for the specified account within the selected date range.
EndDate	Yes	google.protobuf.Timestamp EndDate	Indicate the end date for your order search query. Response to this query contains orders till the EndDate
EndTime	No	google.protobuf.Duration EndTime	Provide the end time for your order search query. In case none is given, by default EndTime is taken as "23:59:59"
IncludeExchangeTradeOrder	No	bool	Boolean flag to filter the order responses based on order type ExchangeTradeOrder

Field Name	Required?	Data Type	Accepted Values/Examples
OriginalOrderId	No	string	Original order identification code. Note: Unlike other search fields, this field can be used independently and does not require any additional search criteria, including **Start Date** or **End Date**.
StartDate	Yes	google.protobuf.Timestamp StartDate	Indicate the start date for your order search query. Response to this query contains orders originating from StartDate
StartTime	No	google.protobuf.Duration StartTime	Provide the start time for your order search query. In case none is given, by default StartTime is taken as "00:00:00"
Symbol	No	string	Valid ticker symbol (e.g., AAPL, IBM, VOD.LSE). Note: Use this field with the **Start Date** and **End Date** fields to retrieve order details for a specific symbol within the selected date range. Optionally, include **Account** details to further refine the search.
TimeoutInSeconds	No	int32	Time out interval for the API request
UserToken	Yes	string	A server generated GUID that is given as response to the client during the first login

OrderDetailByDateRangeResponse

Field Name	Data Type	Retrieves
Acknowledgement	ServerAcknowledgement	Response from the server (success or failure) Refer to the section ServerAcknowledgement
OrderDetail	string	OrderInfo object is retrieved in JSON format

Note: The GetOrderDetailByDateRange API allows you to fetch historical data (up to 3 years) for user-submitted order events including USO, USSOs, FEs, and ETOs. The date range per request cannot exceed 1 year. Start/End dates cannot be greater than the current date.

GetOrderDetailByOrderId

The **OrderDetailByOrderIdRequest** object is used as an input in **GetOrderDetailByOrderId** API. This API is used to fetch order details during the day by providing order ID.

OrderDetailByOrderIdRequest

Field Name	Required?	Data Type	Accepted Values/Examples
OrderId	Yes	string	A unique id associated to every order
UserToken	Yes	string	A server generated GUID that is given as response to the client during the first login

OrderDetailByOrderIdResponse

Field Name	Data Type	Retrieves
OrderDetails	repeated Order-DetailsResponse	Refer to the section OrderDetailsResponse
Acknowledgement	ServerAcknowledgement	Response from the server (success or failure) Refer to the section Server-Acknowledgement

GetOrderDetailByOrderTag

The **OrderDetailByOrderTagRequest** object is used as an input in **GetOrderDetailByOrderTag** API. This API allows you to fetch the Order ID of an order using the order tag and simultaneously fetch the order

activity (submit, change, cancel etc.) during the day. You can retrieve the details of multiple orders by sending multiple order tags along with their order event type.

OrderDetailByOrderTagRequest

Field Name	Required?	Data Type	Accepted Values/Examples
OrderEventType	Yes	string	Order event type (e.g., UserSubmitOrder, User-SubmitStagedOrder, User-SubmitCompoundOrder)
OrderTags	Yes	repeated string	Order tag
UserToken	Yes	string	A server generated GUID that is given as response to the client during the first login

OrderDetailByOrderTagResponse

Field Name	Data Type	Retrieves
Acknowledgement	ServerAcknowledgement	Response from the server (success or fail-ure) Refer to the section Server-Acknowledgement
OrderDetails	repeated Order-DetailsResponse	OrderInfo object is retrieved in JSON format

GetUserAccounts

The **UserAccountsRequest** object is used as an input in **GetUserAccounts** API. This API is used to fetch all the user accounts linked to your account in Eze EMS.

UserAccountsRequest

Field Name	Required?	Data Type	Accepted Values/Examples
UserToken	Yes	string	A server generated GUID that is given as response to the client during the first login

UserAccountsResponse

Field Name	Data Type	Retrieves
Accounts	map<string, string>	This provides a list of all existing user accounts for the requested client. Example: <Account0, “TAL;TEST;USER1;TRADE”><Account1, “TAL;TEST;USER2;NEUTRAL”>
Acknowledgement	ServerAcknowledgement	Response from the server (success or failure). Refer to the section ServerAcknowledgement

SubmitAllocationOrder

The **SubmitAllocationOrderRequest** object is used as an input in **SubmitAllocationOrder** API. This API is used to allocate partial or full order volume from your account to multiple sub-accounts by sending allocation instructions over Allocations API.

SubmitAllocationOrderRequest

Field Name	Required?	Data Type	Accepted Values/Examples
Exchange	Yes	string	Three-letter acronym for listing exchange
OrderId	Yes	string	A unique id associated to every order. This is the identifier to lookup the specific order to allocate
OrderTypes	Yes	OrderType	Refer to the section OrderType
SourceAccount	Yes	string	Source account for the allocated order
Symbol	Yes	string	Valid ticker symbol (e.g., AAPL, IBM, VOD.LSE)
TargetAccount	Yes	string	Destination account for the allocated order
TargetPrice	No	google.protobuf.DoubleValue	Limit price of allocated order

Field Name	Required?	Data Type	Accepted Values/Examples
TargetQuantity	Yes	int32	Volume of the ticket if the ticket is live, or the Traded Amount if the order has been canceled before completion
TypeOfAllocation	Yes	AllocationType	Refer to the section AllocationType
UserToken	Yes	string	A server generated GUID that is given as response to the client during the first login

SubmitAllocationOrderResponse

Field Name	Data Type	Retrieves
Acknowledgement	ServerAcknowledgement	Response from the server (success or failure). Refer to the section ServerAcknowledgement
OptionalFields	map<string, string>	This is additional information given to the client in cases of failure. The field OptionalFields ["ErrorMessage"] would contain the reason for error or exception
ServerResponse	string	Success: In case the invoked API executes successfully on the server side Failed: In case the invoked API throws an error or lands in an exception

SubmitBasketOrder

The **BasketOrderRequest** object is used as an input in **SubmitBasketOrder** API. Using this API you can stage and execute a basket of orders in EMS xAPI.

BasketOrderRequest

Field Name	Required?	Data Type	Accepted Values/Examples
Orders	Yes	repeated OrderRow	Refer to the section OrderRow

Field Name	Required?	Data Type	Accepted Values/Examples
Styp	Yes	enum	Security Type (extended field): 1 = Equities, 2 = Options, 3 = Futures. Defaults to 1 (Equities) if omitted. For futures and options, this field is required; otherwise, the order is treated as an equity order.
UserToken	Yes	string	A server generated GUID that is given as response to the client during the first login

BasketOrderResponse

Field Name	Data Type	Retrieves
Acknowledgement	ServerAcknowledgement	Response from the server (success or failure). Refer to the section ServerAcknowledgement
OptionalFields	map<string, string>	This is additional information given to the client in cases of failure. The field OptionalFields ["ErrorMessage"] would contain the reason for error or exception
ServerResponse	string	Success: In case the invoked API executes successfully on the server side Failed: In case the invoked API throws an error or lands in an exception

SubmitBookTrade

The **SubmitBookTradeRequest** object is used as an input in **SubmitBookTrade** API. This API can be used to fetch post trade details (e.g., commission, commission type, allocations, allocated quantity etc.) using the order ID.

SubmitBookTradeRequest

Field Name	Required?	Data Type	Accepted Values/Examples
AllocDetails	Yes	repeated AllocationDetails	Refer to the section AllocationDetails
OrderId	Yes	string	A unique ID associated with each order. This is the identifier to lookup the specific order to modify
OrderTag	Yes	string	Order tag

Field Name	Required?	Data Type	Accepted Values/Examples
SourceAccount	Yes	string	Source account for the allocated order
UserToken	Yes	string	A server generated GUID that is given as response to the client during the first login

SubmitBookTradeResponse

Field Name	Data Type	Retrieves
Acknowledgement	ServerAcknowledgement	Response from the server (success or failure). Refer to the section ServerAcknowledgement
OptionalFields	map<string, string>	This is additional information given to the client in cases of failure. The field OptionalFields ["ErrorMessage"] would contain the reason for error or exception
ServerResponse	string	Success: In case the invoked API executes successfully on the server side Failed: In case the invoked API throws an error or lands in an exception

SubmitPairOrder

The **SubmitPairOrderRequest** object is used as an input in **SubmitPairOrder** API. This API is used to place a broker specific pair algo order. A Pair order is a multi-leg (two) order, these legs are sent out to the exchange or broker as one unit.

SubmitPairOrderRequest

Field Name	Required?	Data Type	Accepted Values/Examples
Account	Yes	string	Semi-colon separated values that represent either Trade or Neutral accounts the user has permission to e.g., TAL;TEST;USER1;TRADE or TAL;TEST;USER2;NEUTRAL
ClaimRequire	No	bool	TRUE or FALSE (Note: setting TRUE value envisages a user running Eze EMS who then claims the Order so it can switch from Pending to Live State)
Leg1_Quantity	Yes	int32	Value > 0

Field Name	Required?	Data Type	Accepted Values/Examples
Leg1_Side	Yes	string	BUY, SELL, SELLSHORT
Leg1_Symbol	Yes	string	Valid ticker symbol (e.g., AAPL, IBM, VOD.LSE)
Leg2_Quantity	Yes	int32	Value > 0
Leg2_Side	Yes	string	BUY, SELL, SELLSHORT
Leg2_Symbol	Yes	string	Valid ticker symbol (e.g., AAPL, IBM, VOD.LSE)
OrderTag	No	string	Order tag
Route	Yes	string	Route name as shown in Eze EMS. Note: This field is also referred to as **Exit Vehicle**
Staged	No	bool	TRUE or FALSE (Note: in order to send a staged order, this field becomes mandatory and has to be set as TRUE only)
Styp	Yes	enum	Security Type (extended field): 1 = Equities, 2 = Options, 3 = Futures. Defaults to 1 (Equities) if omitted. For futures and options, this field is required; otherwise, the order is treated as an equity order.
TicketId	No	string	Ticket ID
UserToken	Yes	string	A server generated GUID that is given as response to the client during the first login

SubmitPairOrderResponse

Field Name	Data Type	Retrieves
Acknowledgement	ServerAcknowledgement	Response from the server (success or failure). Refer to the section ServerAcknowledgement
OptionalFields	map<string, string>	This is additional information given to the client in cases of failure. The field OptionalFields ["ErrorMessage"] would contain the reason for error or exception
ServerResponse	string	Success: In case the invoked API executes successfully on the server side Failed: In case the invoked API throws an error or lands in an exception

SubmitSeedData

The **SubmitSeedDataRequest** object is used as an input in **SubmitSeedData** API. This API is used to validate and synchronize the symbol with exchange and seed the symbol with your account initially, when you are trying to use a symbol for the first time on your account.

SubmitSeedDataRequest

Field Name	Required?	Data Type	Accepted Values/Examples
Account	Yes	string	Semi-colon separated values that represent either Trade or Neutral accounts the user has permission to e.g., TAL;TEST;USER1;TRADE or TAL;TEST;USER2;NEUTRAL
Symbol	Yes	string	Valid ticker symbol (e.g., AAPL, IBM, VOD.LSE)
UserToken	Yes	string	A server generated GUID that is given as response to the client during the first login

SubmitSeedDataResponse

Field Name	Data Type	Retrieves
Acknowledgement	ServerAcknowledgement	Response from the server (success or failure). Refer to the section ServerAcknowledgement
OptionalFields	map<string, string>	This is additional information given to the client in cases of failure. The field OptionalFields [“ErrorMessage”] would contain the reason for error or exception
ServerResponse	string	Success: In case the invoked API executes successfully on the server side Failed: In case the invoked API throws an error or lands in an exception

SubmitSingleOrder

The **SubmitSingleOrderRequest** object is used as an input in **SubmitSingleOrder** API. This API is used to place a single leg buy/sell order (either staged or live).

SubmitSingleOrderRequest

Field Name	Required?	Data Type	Accepted Values/Examples
Account	Yes	string	Semi-colon separated values that represent either Trade or Neutral accounts the user has permission to e.g., TAL;TEST;USER1;TRADE or TAL;TEST;USER2;NEUTRAL
ClaimRequire	No	bool	TRUE or FALSE (Note: setting TRUE value envisages a user running Eze EMS who then claims the Order so it can switch from Pending to Live State)
ExpirationDate	No	google.protobuf.Timestamp	Date at which order is no longer valid
GoodFrom	No	google.protobuf.Timestamp	Time at which order is first valid for execution
OrderTag	No	string	Order tag
Price	No	google.protobuf.DoubleValue	Limit price submitted in order
PriceType	No	PriceTypeEnum	By default the price type is set to **Market**, in case no value is provided. Refer to the section PriceTypeEnum
Quantity	Yes	int32	Value > 0
ReturnResult	No	bool	If set to **True**, a current snapshot of all the order event fields is retrieved
ReturnResultTimeOutInMs	No	int32	If set to **True**, fetches the results at the defined time out interval in milli seconds. If set to **True**, but no specific value is defined, the default server value (e.g., "RequestTimeOutLimit": 10000) is used.

Field Name	Required?	Data Type	Accepted Values/Examples
Route	Yes	string	Route name as shown in Eze EMS. Note: This field is also referred to as **Exit Vehicle**
Side	Yes	string	BUY, SELL, or SELLSHORT. To send a sell short order, assign the SHORT_LOCATE_ID extended field. To send a buy to cover order, set the side to BUY and assign the TO_OPEN_POS extended field: • 100: Open a new position • 101: Close a position • When omitted, the system automatically determines position intent based on Order side (buy or sell), User account trading permissions, Existing positions
Staged	No	bool	TRUE or FALSE Note: In order to send a staged order, this field becomes mandatory and has to be set as TRUE only
StopPrice	No	google.protobuf.DoubleValue	Stop price
Styp	Yes	enum	Security Type (extended field): 1 = Equities, 2 = Options, 3 = Futures. Defaults to 1 (Equities) if omitted. For futures and options, this field is required; otherwise, the order is treated as an equity order.
Symbol	Yes	string	Valid ticker symbol (e.g., AAPL, IBM, VOD.LSE)

Field Name	Required?	Data Type	Accepted Values/Examples
TicketId	No	string	Ticket ID
TimeInForce	No	ExpirationType	Time or date at which order is no longer valid. In case no value is provided, DAY expiration type is set by default. Refer to the section ExpirationType
UserMessage	No	string	User message/notes
UserToken	Yes	string	A server generated GUID that is given as response to the client during the first login

SubmitSingleOrderResponse

Field Name	Data Type	Retrieves
Acknowledgement	ServerAcknowledgement	Response from the server (success or failure). Refer to the section ServerAcknowledgement
OptionalFields	map<string, string>	This is additional information given to the client in cases of fail-ure. The field OptionalFields ["ErrorMessage"] would contain the reason for error or exception
OrderDetails	OrderDetailsResponse	Refer to the section OrderDetailsResponse
ServerResponse	string	Success: In case the invoked API executes successfully on the server side Failed: In case the invoked API throws an error or lands in an exception

SubmitTradeReport

The **SubmitTradeReportRequest** object is used as an input in **SubmitTradeReport** API. This API allows you to update execution and fills info of an orders manually. Only the orders with **Foreign Execution** flag are reported to exchange/clearing firm.

SubmitTradeReportRequest

Field Name	Required?	Data Type	Accepted Values/Examples
Account	Yes	string	Semi-colon separated values that represent either Trade or Neutral accounts the user has permission to e.g., TAL;TEST;USER1;TRADE or TAL;TEST;USER2;NEUTRAL
OrderType	Yes	TradeType	Refer to the section TradeType
Quantity	Yes	int32	Value > 0
Route	Yes	string	Route name as shown in Eze EMS. Note: This field is also referred to as **Exit Vehicle**
Side	Yes	string	BUY, SELL, or SELLSHORT. To send a sell short order, assign the SHORT_LOCATE_ID extended field. To send a buy to cover order, set the side to BUY and assign the TO_OPEN_POS extended field: • 100: Open a new position • 101: Close a position • When omitted, the system automatically determines position intent based on Order side (buy or sell), User account trading permissions, Existing positions
Styp	Yes	enum	Security Type (extended field): 1 = Equities, 2 = Options, 3 = Futures. Defaults to 1 (Equities) if omitted. For futures and options, this field is required; otherwise, the order is treated as an equity order.
Symbol	Yes	string	Valid ticker symbol (e.g., AAPL, IBM, VOD.LSE)
UserToken	Yes	string	A server generated GUID that is given as response to the client during the first login

SubmitTradeReportResponse

Field Name	Data Type	Retrieves
Acknowledgement	ServerAcknowledgement	Response from the server (success or failure). Refer to the section ServerAcknowledgement

Field Name	Data Type	Retrieves
OptionalFields	map<string, string>	This is additional information given to the client in cases of failure. The field OptionalFields ["ErrorMessage"] would contain the reason for error or exception
ServerResponse	string	Success: In case the invoked API executes successfully on the server side Failed: In case the invoked API throws an error or lands in an exception

SubscribeOrderInfo

The **SubscribeOrderInfoRequest** object is used as an input in **SubscribeOrderInfo** API. This API is used to fetch the order activity (submit, change, cancel etc.) during the day.

SubscribeOrderInfoRequest

Field Name	Required?	Data Type	Accepted Values/Examples
ExcludeHistory	No	bool	Boolean flag to exclude the historic order responses and return only the latest events
FieldSelection	No	enum	Field selection
IncludeClerkReject	No	bool	Boolean flag to filter the order responses based on order type ClerkReject
IncludeExchangeAcceptOrder	No	bool	Boolean flag to filter the order responses based on order type ExchangeAcceptOrder
IncludeExchangeTradeOrder	No	bool	Boolean flag to filter the order responses based on order type ExchangeTradeOrder
IncludeForeignExecution	No	bool	Boolean Flag to filter the order responses based on order type ForeignExecution
IncludeUserSubmitAllocation	No	bool	Boolean flag to filter the order responses based on order type UserSubmitAllocation

Field Name	Required?	Data Type	Accepted Values/Examples
IncludeUserSubmitAllocationEx	No	bool	Boolean flag to filter the order responses based on order type UserSubmitAllocationEx
IncludeUserSubmitCancel	No	bool	Boolean flag to filter the order responses based on order type UserSubmitCancel
IncludeUserSubmitChange	No	bool	Boolean flag to filter the order responses based on order type UserSubmitChange
IncludeUserSubmitCompoundOrder	No	bool	Boolean flag to filter the order responses based on order type UserSubmitCompoundOrder
IncludeUserSubmitOrder	No	bool	Boolean flag to filter the order responses based on order type UserSubmitOrder
IncludeUserSubmitStagedOrder	No	bool	Boolean flag to filter the order responses based on order type UserSubmitStagedOrder
IncludeUserSubmitTradeReport	No	bool	Boolean flag to filter the order responses based on order type UserSubmitTradeReport
SubscriptionLevel	No	enum	FILL = 0; STATUS = 1 (This Field is deprecated)
UserToken	Yes	string	A server generated GUID that is given as response to the client during the first login

SubscribeOrderInfoResponse

Field Name	Data Type	Retrieves
Account	string	Semi-colon separated values that represent either Trade or Neutral accounts the user has permission to e.g., TAL;TEST;USER1;TRADE or TAL;TEST;USER2;NEUTRAL

Field Name	Data Type	Retrieves
Acknowledgement	ServerAcknowledgement	Response from the server (success or failure). Refer to the section Server-Acknowledgement
Buyorsell	string	BUY, SELL, or SELLSHORT. This field will soon be deprecated. Alternatively, you can use the field **Side**
ClaimedByClerk	string	Claimed by clerk
CurrentStatus	string	Current status of order (PENDING, LIVE, COMPLETED, or DELETED)
ExpirationDate	google.protobuf.Timestamp	Date at which order is no longer valid
GoodFrom	google.protobuf.Timestamp	Time at which order is first valid for execution
LinkedOrderId	string	ID of linked order
OrderId	string	A unique id associated to every order. This is the identifier to lookup the specific order to cancel
OrderTag	string	Order tag
OriginalOrderId	string	Original order identification code
PairSpreadType	string	Pair spread type
Price	google.protobuf.DoubleValue	Limit price submitted in order
PriceType	PriceTypeEnum	By default the price type is set to **Market**, in case no value is provided. Refer to the section PriceTypeEnum
Reason	string	Reason given by user or destination for changing, cancelling, or deleting the order
RefersToId	string	Refers to the identification code
Route	string	Route name as shown in Eze EMS. Note: This field is also referred to as **Exit Vehicle**

Field Name	Data Type	Retrieves
Side	string	BUY, SELL, or SELLSHORT. To send a sell short order, assign the SHORT_LOCATE_ID extended field. To send a buy to cover order, set the side to BUY and assign the TO_OPEN_POS extended field: • 100: Open a new position • 101: Close a position • When omitted, the system automatically determines position intent based on Order side (buy or sell), User account trading permissions, Existing positions
StopPrice	google.protobuf.DoubleValue	Stop price
submitTime	google.protobuf.Timestamp	The time when the trade was submitted
Symbol	string	Valid ticker symbol (e.g., AAPL, IBM, VOD.LSE)
TicketId	string	Ticket ID
TimeInForce	ExpirationType	Time or date at which order is no longer valid. Refer to the section ExpirationType
TimeStamp	string	Timestamp applied to order by trading system
TraderId	string	USERNAME@DOMAIN of the message recipient
Type	string	Order Event Type. Refer to the section Appendix A
UserMessage	string	User message/notes
Volume	int32	Order quantity
VolumeTraded	int32	Volume traded so far by the order

The following table provides you the Extended Fields details for Data Type **map<string, string>** used in SubscribeOrderInfoResponse.

Key	Expected Values
AcctType	Type of account for a position
AllocatedValue	Allocated value
Ask	Last ask price for symbol
AvgPrice	Average execution price for order so far
Basisvalue	Currency value of a unit move
Bid	Last bid price for symbol
BLOOMBERG_CODE	Bloomberg symbol code associated with the order
BLOOMBERG_CODE_FULL	Full bloomberg symbol code
BookId	Book ID
BOOKING_TYPE	Booking type
ClientOrderId	Client order ID
Commission	Net of commissions for today's trades for a symbol
CommissionCode	Broker code
CommissionRate	Commission rate
CommissionRateType	Commission rate type
CrossFlag	Cross flag
Currency	Currency of symbol being traded or Currency of position
DateIndex	Date index
EcnFee	ECN fee
Exchange	Three-letter acronym for listing exchange
ExecutionState	Execution state
ExtendedStateFlags	Extended state information for internal use
ExtendedStateFlags2	Extended state flags 2
EXTERNAL_CUSTOMER	External customer
EXTERNAL_TRADER	External trader

Key	Expected Values
ExternalAcceptanceFlag	External acceptance flag
FixTraderId	FIX trader ID
FornexSourceFlags	Forex source
FREE_QTY	Target quantity - Committed quantity (the amount of the ticket that has not filled and is not in the market)
GwBookSeqNo	Gateway book sequence number
Latency3	Latency
Latency6	Latency
LinkedOrderCancellation	Linked order cancellation
LinkedOrderRelationship	Linked order relationship
Minmove	Minimum movement
NewRemoteId	New remote identification code
NewsDate	Date of most recent news story
NewsTime	Time of most recent news story
OmsClientType	OMS client type
OppositeParty	Contra (used by ARCAEX to signify liquidity added or removed)
OrderFlags	Order flags
OrderFlags2	Order flag 2
OrderResidual	Residual volume
OriginalPrice	Original price
OriginalTraderId	USERNAME@DOMAIN of the trader who placed the trade originally
OriginalVolume	Original volume of trade
PairCash	Pair cash component
PairImbalanceLimit	Pair imbalance limit

Key	Expected Values
PairImbalanceLimitType	Pair imbalance limit type
PairLeg1Benchmark	Pair Leg 1 benchmark
PairLeg1BenchmarkType	Pair Leg 1 benchmark
PairLeg2Benchmark	Pair Leg 2 benchmark
PairLeg2BenchmarkType	Pair Leg 2 benchmark
PairRatio	Pair ratio
PairSpread	Pair spread
PairTarget	Pair target
Putcallind	Option type (P = put, C = call, U = underlier) for symbol
Rank	Rank
REGION	Region associated with the listed exchanges for a symbol
RemainingVolume	Remaining volume
RemoteId	Remote ID
RIC_CODE	Reuters Instrument Code
SERVER_ARRIVAL_PRICE	Arrival price as stamped by server. Will be 0 when order arrives outside core trading hours
SharesAllocated	Shares allocated
ShortLocateId	ID assigned to short sell orders as required by regulation SHO
SpreadClip	Spread clip
SpreadClipType	Spread clip type
SpreadLegCount	Number of legs this spread contains
SpreadLegLeanPriority	Spread leg lean priority
SpreadLegNumber	Number of legs this spread contains
SpreadLegPriceType	Spread leg price type
SpreadNumLegs	Spread number of legs

Key	Expected Values
StrikePrc	Price at which option can be exercised
Styp	Security type
Table	Table
TboAccountId	Account ID
TraderCapacity	Trader capacity
TRD_TIME	Time of trade
TrdTime	Time of last trade
Undersym	Underlier symbol
UtcOffset	UTC offset
VolumeType	Indicates the fill type of the position for the symbol
WorkingQty	Quantity out in the market

SubscribeOrderInfoJson

The **SubscribeOrderInfoJsonRequest** object is used as an input in **SubscribeOrderInfoJson** API. This API can stream order updates in JSON format and helps in building the DataFrame row by row.

SubscribeOrderInfoJsonRequest

Field Name	Required?	Data Type	Accepted Values/Examples
Filters	Yes	OrderInfoFilters	Refer to the section OrderInfoFilters
UserToken	Yes	string	A server generated GUID that is given as response to the client during the first login

SubscribeOrderInfoJsonResponse

Field Name	Data Type	Retrieves
Acknowledgement	ServerAcknowledgement	Response from the server (success or failure) Refer to the section ServerAcknowledgement
OrderInfoJson	string	OrderInfo object is retrieved in JSON format

Utility APIs

The following APIs are provided as part of Utility:

API Name	Input	Output
ChangePassword	ChangePasswordRequest	ChangePasswordResponse
ChangePasswordSRP	ChangePasswordSRPRequest	ChangePasswordSRPResponse
CompleteLoginSrp	CompleteLoginSrpRequest	CompleteLoginSrpResponse
Connect	ConnectRequest	ConnectResponse
CreateStrategy	CreateStrategyRequest	CreateStrategyResponse
Disconnect	DisconnectRequest	DisconnectResponse
GetFirmName	GetFirmNameRequest	GetFirmNameResponse
GetStrategyData	GetStrategyDataRequest	GetStrategyDataResponse
GetStrategyList	StrategyListRequest	StrategyListResponse
GetTodaysActivity	TodaysActivityRequest	TodaysActivityResponse
GetTodaysActivityJson	TodaysActivityJsonRequest	TodaysActivityJsonResponse
GetTodaysBalances	TodaysBalancesRequest	TodaysBalancesResponse
GetTodaysBrokenDownPos-itions	TodaysBrokenDownPos-itionsRequest	TodaysBrokenDownPos-itionsResponse
GetTodaysNetPositions	TodaysNetPositionsRequest	TodaysNetPositionsResponse
GetUserRouteProps	GetUserRoutePropsRequest	GetUserRoutePropsResponse
GetUserRoutes	GetUserRoutesRequest	GetUserRoutesResponse
MultiFactorAuthentication	MultiFact-orAuthenticationRequest	MultiFact-orAuthenticationResponse
StartLoginSrp	StartLoginSrpRequest	StartLoginSrpResponse
SubscribeHeartBeat	SubscribeHeartBeatRequest	SubscribeHeartBeatResponse
SubscribeTodaysNetPositions	SubscribeNetPositionsRequest	stream Sub-scribeNetPositionsResponse

API Name	Input	Output
UnSub-scribeTodaysNetPositions	UnSub-scribeNetPositionsRequest	UnSub-scribeNetPositionsResponse
UpdateStrategy	UpdateStrategyRequest	UpdateStrategyResponse
UserExists	UserExistsRequest	UserExistsResponse

Custom Data Type

AggregatePositionRecord

Field Name	Data Type	Accepted Values/Examples
SoldShortQty	double	Total number of short shares sold for the symbol
Bank	string	Bank of trading account for selected position
BoughtAvgPrc	double	Average price shares bought
BoughtQty	double	Bought order quantity
Branch	string	Branch of trading account for selected position
Customer	string	Customer name of trading account for selected position
Deposit	string	Deposit or account name for selected position
DispName	string	'IBM' (Symbol formats are datafeed-specific. Check your feed handler help file for the correct format.)
ExtendedFields	map<string, string>	Extended fields. Refer to the section in OrderResponse
IntradayAvgPrc	double	Intraday average price
IntradayLongAvgPrc	double	Intraday average price long
IntradayLongPos	double	Intraday long position
IntradayPos	double	Intraday position
IntradayShortAvgPrc	double	Average intraday execution price for short order
IntradayShortPos	double	Intraday short position
LongAvgPrc	double	Average price long

Field Name	Data Type	Accepted Values/Examples
LongPos	double	Long position
ShortAvgPrc	double	Short average price
ShortPos	double	Short position
SoldAvgPrc	double	Sold average price
SoldQty	double	Sold order quantity
SoldShortAvgPrc	double	Sold short average price
TotalAvgPrc	double	Total average execution price for order so far
TotalPos	double	Total position

BookOrder

Following is the structure of "BookOrder".

Field Name	Data Type	Accepted Values/Examples
Allocations	TodaysActivityResponse	Refer to the section TodaysActivityResponse
AveragePrice	double	Average execution price for order so far
CancelOrders	TodaysActivityResponse	Refer to the section TodaysActivityResponse
ChangeOrders	TodaysActivityResponse	Refer to the section TodaysActivityResponse
ExecutedVolume	int64	Executed volume
Executions	TodaysActivityResponse	Refer to the section TodaysActivityResponse
IsChangeOrCancelPending	bool	Is change or cancel pending
Order	OrderResponse	Refer to the section OrderResponse
OrderHistory	TodaysActivityResponse	Refer to the section TodaysActivityResponse
RemainingVolume	int64	Remaining volume

DepositRow

Field Name	Data Type	Accepted Values/Examples
Bank	string	Bank of trading account for selected position
Branch	string	Branch of trading account for selected position
CashBalance	double	Cash balance
CashBalanceAdj	double	Cash balance adjustment
CbpRemaining	double	Available funds (Funds - Margin requirements)
Commission	double	Net of commissions for today's trades for a symbol
Cost	double	Cost
Currency	string	Currency of symbol being traded or Currency of position
Customer	string	Customer name of trading account for selected position
DaysBack	int32	The number of days worth of bars to retrieve
Deposit	string	Deposit or account name for selected position
EquityBalance	double	Equity balance
ExcessEq	double	Excess equity
ExcessEq0	double	Start of day excess equity
ExtendedFields	map<string, string>	Extended fields. Refer to the section in OrderResponse
ExtraCbp	double	Cash credit for the account
HistScalpedProfit	double	Historical realized P&L
HouseExcess0	double	Overnight house excess
Marketvalue0	double	Start of day market value
MaxOrderSize	double	Maximum order size allowed in an account
MinBalance	double	Minimum amount of excess equity to trade
Mmr	double	Maintenance margin required for a position

Field Name	Data Type	Accepted Values/Examples
Mmr0	double	Margin requirement for a overnight position
Mmr0Adj	double	Overnight maintenance margin release amount
Mmr0NonDaytradeAdj	double	Adjustment to overnight maintenance margin requirement
NetFees	double	Sum of all trading fees for a position from today's trades
Orq0	double	Morning Orq
PendingMargin	double	Pending margin for this order
Rank	int32	Rank
ScalpedProfit	double	Realized profit/loss for a position
Sma0	double	Uploaded information for special memorandum account requirement
Sma0Adj	double	Adjustment to overnight SMA requirement
SrvPendingMargin	double	Margin for outstanding trades as calculated by tradeserver. Only available for accounts that are configured for this feature
Table	string	Table
TboAccountId	int32	Account ID

ExpirationType

Field Name	Data Type	Accepted Values/Examples
ExpirationTypes	enum	DAY = 0; GTC = 1; GTX = 2; CLO = 3; OPG = 4; IOC = 5; GTD = 6; OTHER = 7 **Day**: Expiration is set for 24 hours after order submission **GTC**: Good Till Cancel Order. As the name suggests, this type of order will theoretically stay live forever until canceled or filled **GTX**: An order to buy or sell that is canceled prior to the market entering into an auction, or crossing phase. Typically, markets that support continuous trading will have an auction phase at the beginning and sometimes also at the end of trading to match up orders that have been entered into the exchange's order book during the pre- or post-trading phase (i.e. where no continuous trading was available). A GTX order automatically expires immediately prior to the commencement of a crossing session, i.e. the party originating the order wants to make sure it gets filled during the current continuous auction, and any remaining open quantity should be discarded at the end of the current continuous auction period **CLO**: Closing order. Commonly known as MOC i.e., a non-limit market order, which is executed as near to the closing price as possible- either exactly at, or slightly after the market close **OPG**: Opening order. Commonly known as MOO i.e., an order that will be executed at the day's opening price. **IOC**: A market or limit-price order that is to be executed in whole or in part as soon as it is placed. Any portion not executed immediately will be cancelled **GTD**: Good Till Date order. User must specify the expiration date in the field **ExpirationDate** in order to place a GTD order **OTHER**: Any expiry type that is not mentioned above, but is present on EMS

OrderInfo

Field Name	Data Type	Retrieves
Account	string	Semi colon separated values that represent either Trade or Neutral accounts the user has permission to e.g., TAL;TEST;USER1;TRADE or TAL;TEST;USER2;NEUTRAL
AccountType	int	Type of account used in trade (cash, long, or short)
AveragePrice	double	Average execution price for order so far
ClaimedByClerk	string	Claimed by clerk
Currency	string	Currency of symbol being traded or Currency of position
CurrentStatus	string	Current status of order (PENDING, LIVE, COMPLETED, or DELETED)
Exchange	string	Three-letter acronym for listing exchange
ExpirationDate	DateTime?	Date at which order is no longer valid
GoodFrom	string	Time at which order is first valid for execution
LinkedOrderId	string	ID of linked order
OptionType	string	Option type (Put/Call) for symbol
OrderEventType	string	Order event type (e.g., UserSubmitOrder, User-SubmitStagedOrder, UserSubmitCompoundOrder)
OrderId	string	A unique id associated to every order. This is the identifier to lookup the specific order to cancel
OrderTag	string	Order tag
OriginalOrderId	string	Original order identification code
Price	decimal	Limit price submitted in order
PriceType	PriceTypeEnum	By default the price type is set to **Market**, in case no value is provided. Refer to the section PriceTypeEnum
Reason	string	Reason given by user or destination for changing, cancelling, or deleting the order
RefersToId	string	Refers to the identification code

Field Name	Data Type	Retrieves
Route	string	Route name as shown in Eze EMS. Note: This field is also referred to as **Exit Vehicle**
SecurityType	int	Security type for the symbol being traded
ShortLocateId	string	ID assigned to short sell orders as required by regulation SHO
Side	string	BUY, SELL, or SELLSHORT. To send a sell short order, assign the SHORT_LOCATE_ID extended field. To send a buy to cover order, set the side to BUY and assign the TO_OPEN_POS extended field: • 100: Open a new position • 101: Close a position • When omitted, the system automatically determines position intent based on Order side (buy or sell), User account trading permissions, Existing positions
StopPrice	decimal	Stop price
StrikePrc	decimal	Price at which option can be exercised
Symbol	string	Valid ticker symbol (e.g., AAPL, IBM, VOD.LSE)
TicketId	string	Ticket ID
TimeInForce	ExpirationType	Time at which order is no longer valid. Refer to the section ExpirationType
TimeStamp	string	Timestamp applied to order by trading system
TraderId	string	USERNAME@DOMAIN of the message recipient
TradeTime	TimeSpan	Time of trade
UnderlierSymbol	string	Underlier symbol
UserMessage	string	User message/notes
Volume	int	Order quantity
VolumeTraded	int	Volume traded so far by the order
VolumeType	string	Indicates the fill type of the position for the symbol

OrderResponse

Field Name	Data Type	Retrieves
Account	string	Semi colon separated values that represent either Trade or Neutral accounts the user has permission to e.g., TAL;TEST;USER1;TRADE or TAL;TEST;USER2;NEUTRAL
Buyorsell	string	BUY, SELL, or SELLSHORT. This field will soon be deprecated. Alternatively, you can use the field **Side**
ClaimedByClerk	string	Claimed by clerk
CurrentStatus	string	Current status of order (e.g., PENDING, LIVE, COMPLETED, DELETED)
ExpirationDate	google.protobuf.Timestamp	Date at which order is no longer valid
GoodFrom	google.protobuf.Timestamp	Time at which order is first valid for execution
LinkedOrderId	string	ID of linked order
OrderId	string	A unique id associated to every order. This is the identifier to lookup the specific order to cancel
OrderTag	string	Order tag
OriginalOrderId	string	Original order identification code
PairSpreadType	string	Pair spread type
Price	google.protobuf.DoubleValue	Limit price submitted in order
PriceType	PriceTypeEnum	By default the price type is set to **Market**, in case no value is provided. Refer to the section PriceTypeEnum
Reason	string	Reason given by user or destination for changing, cancelling, or deleting the order
RefersToId	string	Refers to identification code
Route	string	Route name as shown in Eze EMS. Note: This field is also referred to as **Exit Vehicle**

Field Name	Data Type	Retrieves
Side	string	BUY, SELL, or SELLSHORT. To send a sell short order, assign the SHORT_LOCATE_ID extended field. To send a buy to cover order, set the side to BUY and assign the TO_OPEN_POS extended field: • 100: Open a new position • 101: Close a position • When omitted, the system automatically determines position intent based on Order side (buy or sell), User account trading permissions, Existing positions
StopPrice	google.protobuf.DoubleValue	Stop price
submitTime	google.protobuf.Timestamp	The time when the trade was submitted
Symbol	string	Valid ticker symbol (e.g., AAPL, IBM, VOD.LSE)
TicketId	string	Ticket ID
TimeInForce	ExpirationType	Time or date at which order is no longer valid. Refer to the section ExpirationType
TimeStamp	string	Timestamp applied to order by trading system
TraderId	string	USERNAME@DOMAIN of the message recipient
Type	string	Order Event Type. Refer to the section Appendix A
UserMessage	string	User message/notes
Volume	int32	Order quantity
VolumeTraded	int32	Volume traded so far by the order

The following table provides you the Extended Fields details for Data Type **map<string, string>** used in OrderResponse.

Key	Expected Values
AcctType	Type of account for a position

Key	Expected Values
AllocatedValue	Allocated value
Ask	Last ask price for symbol
AvgPrice	Average execution price for order so far
Basisvalue	Currency value of a unit move
Bid	Last bid price for symbol
BLOOMBERG_CODE_FULL	Full bloomberg symbol code
BookId	Book ID
BOOKING_TYPE	Booking type
ClientOrderId	Client order ID
Commission	Net of commissions for today's trades for a symbol
CommissionCode	Broker code
CommissionRate	Commission rate
CommissionRateType	Commission rate type
CrossFlag	Cross flag
Currency	Currency of symbol being traded or Currency of position
DateIndex	Date index
EcnFee	ECN fee
Exchange	Three-letter acronym for listing exchange
ExecutionState	Execution state
ExtendedStateFlags	Extended state information for internal use
ExtendedStateFlags2	Extended state flags 2
EXTERNAL_CUSTOMER	External customer
EXTERNAL_TRADER	External trader
ExternalAcceptanceFlag	External acceptance flag
FixTraderId	FIX trader ID

Key	Expected Values
FornexSourceFlags	Forex source
FREE_QTY	Target quantity - Committed quantity (the amount of the ticket that has not filled and is not in the market)
GwBookSeqNo	Gateway book sequence number
Latency3	Latency
Latency6	Latency
LinkedOrderCancellation	Linked order cancellation
LinkedOrderRelationship	Linked order relationship
Minmove	Minimum movement
NewRemoteId	New remote identification code
NewsDate	Date of most recent news story
NewsTime	Time of most recent news story
OmsClientType	OMS client type
OppositeParty	Contra (used by ARCAEX to signify liquidity added or removed)
OrderFlags	Order flags
OrderFlags2	Order flag 2
OrderResidual	Residual volume
OriginalPrice	Original price
OriginalTraderId	USERNAME@DOMAIN of the trader who placed the trade originally
OriginalVolume	Original volume of trade
PairCash	Pair cash component
PairImbalanceLimit	Pair imbalance limit
PairImbalanceLimitType	Pair imbalance limit type
PairLeg1Benchmark	Pair Leg 1 benchmark

Key	Expected Values
PairLeg1BenchmarkType	Pair Leg 1 benchmark
PairLeg2Benchmark	Pair Leg 2 benchmark
PairLeg2BenchmarkType	Pair Leg 2 benchmark
PairRatio	Pair ratio
PairSpread	Pair spread
PairTarget	Pair target
Putcallind	Option type (P = put, C = call, U = underlier) for symbol
Rank	Rank
REGION	Region associated with the listed exchanges for a symbol
RemainingVolume	Remaining volume
RemoteId	Remote ID
RootOrderID	A unique ID associated with each USO child order event, helping you to trace the staged order (USSO)
SERVER_ARRIVAL_PRICE	Arrival price as stamped by server. Will be 0 when order arrives outside core trading hours
SharesAllocated	Shares allocated
ShortLocateId	ID assigned to short sell orders as required by regulation SHO
SpreadClip	Spread clip
SpreadClipType	Spread clip type
SpreadLegCount	Number of legs this spread contains
SpreadLegLeanPriority	Spread leg lean priority
SpreadLegNumber	Number of legs this spread contains
SpreadLegPriceType	Spread leg price type
SpreadNumLegs	Spread number of legs
StrikePrc	Price at which option can be exercised

Key	Expected Values
Styp	Security type
Table	Table
TboAccountId	Account ID
TraderCapacity	Trader capacity
TRD_TIME	Time of trade
TrdTime	Time of last trade
Undersym	Underlier symbol
UtcOffset	UTC offset
VolumeType	Indicates the fill type of the position for the symbol
WorkingQty	Quantity out in the market
WorkingQty	Quantity out in the market

Position

Field Name	Data Type	Accepted Values/Examples
AcctType	int32	Type of account for a position
AverageLong	double	Average price long
AverageLong0	double	The average prize of the overnight long position for the symbol
AverageShort	double	Average price short
AverageShort0	double	The average price of the overnight short position for the symbol
Bank	string	Bank of trading account for selected position
BoughtAvgPrc	double	Average price for order so far
BoughtQty	double	Bought order quantity
Branch	string	Branch of trading account for selected position
Commission	double	Net of commissions for today's trades for a symbol
Currency	string	Currency of symbol being traded or Currency of position

Field Name	Data Type	Accepted Values/Examples
Customer	string	Customer name of trading account for selected position
Deposit	string	Deposit or account name for selected position
DispName	string	'IBM' (Symbol formats are datafeed-specific. Check your feed handler help file for the correct format.)
HistLongPrc0	double	Historic basis price of overnight long position
HistScalpedProfit	double	The sum of hist scalped profit and hist net P&L, plus any commissions and fees for the position
HistShortPrc0	double	Historic basis price of overnight short position
Longpos	double	Long Pposition
Longpos0	double	Remianing overnight long position
Mmr	double	Maintenance margin required for a position
NetFees	double	Sum of all trading fees for a position from today's trades
OrigLongpos0	double	Start of day long position
OrigShortpos0	double	Start of day short position
Rank	int32	Rank
ScalpedProfit	double	Realized profit/loss for a position
Shortpos	double	Short position
Shortpos0	double	Remaining overnight short position
SoldAvgPrc	double	Sold average price
SoldQty	double	Total number of shares sold
SoldShortAvgPrc	double	Average price of Short shares sold for the symbol
SoldShortQty	double	Total number of short shares sold for the symbol
Styp	int32	Security type
Table	string	Table
TboAccountId	int32	Account ID
Undersym	string	Underlier symbol

Price

Field Name	Data Type	Accepted Values/Examples
Base	uint32	Base
BaseCode	int32	Base code
DecimalValue	double	Decimal value of the price
Denominator	int32	Denominator
Int	int32	Integer value of the price
IntegerValue	int32	Integer value
Isnull	bool	Indicates if the price is null or not
Numerator	int32	Numerator

PriceTypeEnum

Field Name	Data Type	Accepted Values/Examples
PriceTypesEnum	enum	Market = 0; Limit = 1; StopMarket = 2; StopLimit = 3; Other = 4; **Market**: An order to buy/sell a security at the most recent market price **Limit**: An order to buy a security at or below a stated price, or to sell a security at or above a stated price **StopMarket**: This order allows the trader to lock in a certain profit and loss (P&L) by deciding a specific range in which a market order can be executed. The stop market order executes a market order immediately when the “stop” price set by a trader is hit by the opposite side of the market (the ask if buying, bid if selling) **StopLimit**: This order declares a “stop” price set by the trader. Once this price is hit by the opposite side of the market, a limit order is immediately placed at a price specified by the trader **Other**: Any price type that is not mentioned above, but is present on EMS

RoutePropsMap

Field Name	Data Type	Accepted Values/Examples
Properties	map<string, string>	Route properties

ServerAcknowledgement

Field Name	Data Type	Retrieves
Message	string	Server response to change password SRP API
OptionalFields	map<string, string>	This is additional information given to the client in cases of failure. The field OptionalFields["ErrorMessage"] would contain the reason for error or exception
ResponseCode	enum	SUCCESS = 0; WARNING = 1; FAILED = 2; ERROR = 3; TIMED_OUT = 4; SESSION_NOT_FOUND = 5
ServerResponse	string	Success: In case the invoked API executes successfully on the server side Failed: In case the invoked API throws an error or lands in an exception

StrategyListRow

Field Name	Data Type	Accepted Values/Examples
AlgoName	string	Algo route
FirmName	string	Firm name associated with the user
StratCreateDate	string	Date of creation of strategy
StrategyName	string	Name of user saved strategy
UserName	string	A valid user name

CreateStrategy

The **CreateStrategyRequest** object is used as an input in **CreateStrategy** API. This API allows you to create new saved strategies.

CreateStrategyRequest

Field Name	Required?	Data Type	Accepted Values/Examples
AlgoName	Yes	string	Algo name

Field Name	Required?	Data Type	Accepted Values/Examples
FirmName	Yes	string	Firm name associated with the user
StrategyName	Yes	string	Name of user saved strategy
StratParams	Yes	string	Strategy parameters
UserToken	Yes	string	A server generated GUID that is given as response to the client during the first login

CreateStrategyResponse

Field Name	Data Type	Retrieves
Acknowledgement	ServerAcknowledgement	Response from the server (success or failure). Refer to the section ServerAcknowledgement

GetFirmName

The **GetFirmNameRequest** object is used as an input in **GetFirmName** API. This API allows you to retrieve the firm name associated with the specified user.

Note: If the firm does not exist, it is created automatically.

GetFirmNameRequest

Field Name	Required?	Data Type	Accepted Values/Examples
UserToken	Yes	string	A server generated GUID that is given as response to the client during the first login

GetUserRoutePropsResponse

Field Name	Data Type	Retrieves
Acknowledgement	ServerAcknowledgement	Response from the server (success or failure). Refer to the section ServerAcknowledgement
FirmName	string	Firm name associated with the user

GetStrategyData

The **GetStrategyDataRequest** object is used as an input in **GetStrategyData** API. This API allows you to retrieve detailed information about a specific saved strategy including the version and binary data. It also provides you the parameter details defined within the strategy.

GetStrategyDataRequest

Field Name	Required?	Data Type	Accepted Values/Examples
AlgoName	Yes	string	Algo name
FirmName	Yes	string	Firm name associated with the user
StrategyName	Yes	string	Strategy name
UserToken	Yes	string	A server generated GUID that is given as response to the client during the first login

GetStrategyDataResponse

Field Name	Data Type	Retrieves
Acknowledgement	ServerAcknowledgement	Response from the server (success or failure). Refer to the section ServerAcknowledgement
StrategyData	StrategyDataRow	Refer to the section StrategyDataRow

GetStrategyList

The **StrategyListRequest** object is used as an input in **GetStrategyList** API. This API is used to fetch the saved stratagies (on Eze EMS) mapped to a broker algo.

StrategyListRequest

Field Name	Required?	Data Type	Accepted Values/Examples
FirmName	Yes	string	Firm name associated with the user
UserToken	Yes	string	A server generated GUID that is given as response to the client during the first login

StrategyListResponse

Field Name	Data Type	Retrieves
Acknowledgement	ServerAcknowledgement	Response from the server (success or failure). Refer to the section ServerAcknowledgement
StrategyList	repeated StrategyListRow	Refer to the section StrategyListRow

GetTodaysActivity

The **TodaysActivityRequest** object is used as an input in **GetTodaysActivity** API. This API is used to fetch your order related details (submit, change, cancel etc.) for the current day.

TodaysActivityRequest

Field Name	Required?	Data Type	Accepted Values/Examples
IncludeClerkReject	No	bool	Boolean flag to filter the order responses based on order type ClerkReject
IncludeExchangeAcceptOrder	No	bool	Boolean flag to filter the order responses based on order type ExchangeAcceptOrder
IncludeExchangeTradeOrder	No	bool	Boolean flag to filter the order responses based on order type ExchangeTradeOrder
IncludeForeignExecution	No	bool	Boolean flag to filter the order responses based on order type ForeignExecution
IncludeOnlyCompleted	No	bool	Boolean flag to filter the order responses based on order event type Completed
IncludeUserSubmitAllocation	No	bool	Boolean flag to filter the order responses based on order type UserSubmitAllocation
IncludeUserSubmitAllocationEx	No	bool	Boolean flag to filter the order responses based on order type UserSubmitAllocationEx
IncludeUserSubmitCancel	No	bool	Boolean flag to filter the order responses based on order type UserSubmitCancel
IncludeUserSubmitChange	No	bool	Boolean flag to filter the order responses based on order type UserSubmitChange

Field Name	Required?	Data Type	Accepted Values/Examples
IncludeUserSubmitCompoundOrder	No	bool	Boolean flag to filter the order responses based on order type UserSubmitCompoundOrder
IncludeUserSubmitOrder	No	bool	Boolean flag to filter the order responses based on order type UserSubmitOrder
IncludeUserSubmitStagedOrder	No	bool	Boolean flag to filter the order responses based on order type UserSubmitStagedOrder
IncludeUserSubmitTradeReport	No	bool	Boolean flag to filter the order responses based on order type UserSubmitTradeReport
TimeoutInSeconds	No	int32	Time out interval for the API request
UserSubmitStagedOrderFullInfo	No	bool	Boolean flag to filter all the child events associated with the parent order
UserToken	Yes	string	A server generated GUID that is given as response to the client during the first login

TodaysActivityResponse

Field Name	Data Type	Retrieves
Acknowledgement	ServerAcknowledgement	Response from the server (success or failure). Refer to the section ServerAcknowledgement
OrderRecordList	repeated OrderResponse	Refer to the section OrderResponse

GetTodaysActivityJson

The **TodaysActivityJsonRequest** object is used as an input in **GetTodaysActivityJson** API. This API provides your today's order activity details in the JSON format.

TodaysActivityJsonRequest

Field Name	Required?	Data Type	Accepted Values/Examples
IncludeClerkReject	No	bool	Boolean flag to filter the order responses based on order type ClerkReject
IncludeExchangeAcceptOrder	No	bool	Boolean Flag to filter the order responses based on order type ExchangeAcceptOrder
IncludeExchangeTradeOrder	No	bool	Boolean flag to filter the order responses based on order type ExchangeTradeOrder
IncludeForeignExecution	No	bool	Boolean flag to filter the order responses based on order type ForeignExecution
IncludeOnlyCompleted	No	bool	Boolean flag to filter the order responses based on order event type Completed
IncludeUserSubmitAllocation	No	bool	Boolean flag to filter the order responses based on order type UserSubmitAllocation
IncludeUserSubmitAllocationEx	No	bool	Boolean flag to filter the order responses based on order type UserSubmitAllocationEx
IncludeUserSubmitCancel	No	bool	Boolean flag to filter the order responses based on order type UserSubmitCancel
IncludeUserSubmitChange	No	bool	Boolean flag to filter the order responses based on order type UserSubmitChange
IncludeUserSubmitCompoundOrder	No	bool	Boolean flag to filter the order responses based on order type UserSubmitCompoundOrder
IncludeUserSubmitOrder	No	bool	Boolean flag to filter the order responses based on order type UserSubmitOrder

Field Name	Required?	Data Type	Accepted Values/Examples
IncludeUserSubmitStagedOrder	No	bool	Boolean flag to filter the order responses based on order type UserSubmitStagedOrder
IncludeUserSubmitTradeReport	No	bool	Boolean flag to filter the order responses based on order type UserSubmitTradeReport
TimeoutInSeconds	No	int32	Time out interval for the API request
UserToken	Yes	string	A server generated GUID that is given as response to the client during the first login

TodaysActivityJsonResponse

Field Name	Data Type	Retrieves
Acknowledgement	ServerAcknowledgement	Response from the server (success or failure). Refer to the section ServerAcknowledgement
TodaysActivityJson	string	OrderInfo object is retrieved in JSON format

GetTodaysBalances

The **TodaysBalancesRequest** object is used as an input in **GetTodaysBalances** API. This API is used to fetch your account balance details (CashBalance, EquityBalance, Margin etc.) for the current day.

TodaysBalancesRequest

Field Name	Required?	Data Type	Accepted Values/Examples
TimeoutInSeconds	No	int32	Time out interval for the API request
UserToken	Yes	string	A server generated GUID that is given as response to the client during the first login

TodaysBalancesResponse

Field Name	Data Type	Retrieves
Acknowledgement	ServerAcknowledgement	Response from the server (success or failure). Refer to the section ServerAcknowledgement

Field Name	Data Type	Retrieves
DepositList	repeated DepositRow	Refer to the section DepositRow

GetTodaysBrokenDownPositions

The **TodaysBrokenDownPositionsRequest** object is used as an input in **GetTodaysBrokenDownPositions** API. This API is used to fetch the broken-down position details for the current day.

TodaysBrokenDownPositionsRequest

Field Name	Required?	Data Type	Accepted Values/Examples
BBCDFilter	No	string	To filter based on a single BBCD level, BBCDFilter parameter must be added to the request before API call. Use semi-colon to separate the BBCD (e.g.; BANK;BRANCH;CUSTOMER;DEPOSIT). Example: To filter for a particular Deposit add BANK;BRANCH;CUSTOMER;DEPOSIT. Similarly, to filter for a particular Customer add BANK;BRANCH;CUSTOMER and BANK;BRANCH or BANK accordingly for Branch or Bank level filter.
TickerFilter	No	string	To filter based on a single or multiple Symbol, TickerFilter parameter must be added to the request before API call. Use semi-colon to add more symbols (e.g.; AAPL;IBM;VOD.LSE).
TimeoutInSeconds	No	int32	Time out interval for the API request
UserToken	Yes	string	A server generated GUID that is given as response to the client during the first login

TodaysBrokenDownPositionsResponse

Field Name	Data Type	Retrieves
Acknowledgement	ServerAcknowledgement	Response from the server (success or failure). Refer to the section ServerAcknowledgement
PositionRecords	repeated Position	Refer to the section Position

GetTodaysNetPositions

The **TodaysNetPositionsRequest** object is used as an input in **GetTodaysNetPositions** API. This API is used to fetch the net position details for the current day.

TodaysNetPositionsRequest

Field Name	Required?	Data Type	Accepted Values/Examples
TimeoutInSeconds	No	int32	Time out interval for the API request
UserToken	Yes	string	A server generated GUID that is given as response to the client during the first login

TodaysNetPositionsResponse

Field Name	Data Type	Retrieves
Acknowledgement	ServerAcknowledgement	Response from the server (success or failure). Refer to the section ServerAcknowledgement
AggregatePositionsList	repeated AggregatePositionRecord	Refer to the section AggregatePositionRecord

GetUserRouteProps

The **GetUserRoutePropsRequest** object is used as an input in **GetUserRouteProps** API. This API allows you to fetch the supported exchanges, route types and security type details for the specified route.

GetUserRoutePropsRequest

Field Name	Required?	Data Type	Accepted Values/Examples
FilterRoutePropsList	No	string	Fetch list of route properties (e.g., VolumeTypes, RouteProp, PriceTypes, ExpirationTypes, BrokerDestination, AllowedToTrade, AlgoType, ATDLType)
FilterRoutesList	No	string	Fetch list of route name(s) as shown in Eze EMS
UserToken	Yes	string	A server generated GUID that is given as response to the client during the first login

GetUserRoutePropsResponse

Field Name	Data Type	Retrieves
Acknowledgement	ServerAcknowledgement	Response from the server (success or failure). Refer to the section ServerAcknowledgement
UserRouteProps	map<string, RoutePropsMap>	Refer to the section RoutePropsMap

GetUserRoutes

The **GetUserRoutesRequest** object is used as an input in **GetUserRoutes** API. This API is used to retrieve route names assigned to a specific user for a given account.

GetUserRoutesRequest

Field Name	Required?	Data Type	Accepted Values/Examples
AccountList	Yes	string	A comma-separated list of accounts, each in BBCD format (e.g., B;B;C;D1, B;B;C;D2)
UserToken	Yes	string	A server generated GUID that is given as response to the client during the first login

GetUserRoutesResponse

Field Name	Data Type	Retrieves
Acknowledgement	ServerAcknowledgement	Response from the server (success or failure). Refer to the section ServerAcknowledgement
UserRoutes	map<string, string>	Refer to the sections AggregatePositionRecord / DepositRow

SubscribeHeartBeat

The **SubscribeHeartBeatRequest** object is used as an input in **SubscribeHeartBeat** API. On subscribing to this API, you will be intimated and requested to reconnect if an active connection with the server is terminated.

SubscribeHeartBeatRequest

Field Name	Required?	Data Type	Accepted Values/Examples
TimeoutInSeconds	No	int32	Time out interval for the API request

Field Name	Required?	Data Type	Accepted Values/Examples
UserToken	Yes	string	A server generated GUID that is given as response to the client during the first login

SubscribeHeartBeatResponse

Field Name	Data Type	Retrieves
Acknowledgement	ServerAcknowledgement	Response from the server (success or failure). Refer to the section ServerAcknowledgement
HeartBeatStatus	enum	LIVE = 0; DEAD = 1; UNKNOWN = 2

SubscribeTodaysNetPositions

The **SubscribeNetPositionsRequest** object is used as an input in **SubscribeTodaysNetPositions** API. By subscribing to this API, you can retrieve real-time position updates through streaming, eliminating the need for repeated API call requests.

SubscribeNetPositionsRequest

Field Name	Required?	Data Type	Accepted Values/Examples
TimeoutInSeconds	No	int32	Time out interval for the API request
UserToken	Yes	string	A server generated GUID that is given as response to the client during the first login

SubscribeNetPositionsResponse

Field Name	Data Type	Retrieves
Acknowledgement	ServerAcknowledgement	Response from the server (success or failure). Refer to the section ServerAcknowledgement
AggregatePositions	AggregatePositionRecord	Refer to the section AggregatePositionRecord

UnSubscribeTodaysNetPositions

The **UnSubscribeNetPositionsRequest** object is used as an input in **UnSubscribeTodaysNetPositions** API. This API is used to unsubscribe the streaming net positions data.

UnSubscribeNetPositionsRequest

Field Name	Required?	Data Type	Accepted Values/Examples
UserToken	Yes	string	A server generated GUID that is given as response to the client during the first login

UnSubscribeNetPositionsResponse

Field Name	Data Type	Retrieves
Acknowledgement	ServerAcknowledgement	Response from the server (success or failure). Refer to the section ServerAcknowledgement

UpdateStrategy

The **UpdateStrategyRequest** object is used as an input in **UpdateStrategy** API. This API allows you to update an existing strategy. The version number is required for updates to prevent conflicts.

UpdateStrategyRequest

Field Name	Required?	Data Type	Accepted Values/Examples
AlgoName	Yes	string	Algo name
FirmName	Yes	string	Firm name associated with the user
StrategyName	Yes	string	Strategy name
StratParams	Yes	string	Strategy parameters
UserToken	Yes	string	A server generated GUID that is given as response to the client during the first login
Version	No	int32	ATDL strategy version number

UpdateStrategyResponse

Field Name	Data Type	Retrieves
Acknowledgement	ServerAcknowledgement	Response from the server (success or failure). Refer to the section ServerAcknowledgement

UserExists

The **UserExistsRequest** object is used as an input in **UserExists** API. This API allows you to checks if a user exists or gets the user's firm.

Note: If the user does not exist, it is created automatically.

UserExistsRequest

Field Name	Required?	Data Type	Accepted Values/Examples
FirmName	Yes	string	Firm name associated with the user
UserToken	Yes	string	A server generated GUID that is given as response to the client during the first login

UserExistsResponse

Field Name	Data Type	Retrieves
Acknowledgement	ServerAcknowledgement	Response from the server (success or failure). Refer to the section ServerAcknowledgement
FirmName	string	The firm name associated with the user
UserExists	bool	Fetches details on if an user exists

Exceptions and Error Handling

For every successful response from the server, the ServerAcknowledgement will yield a ServerResponse called “success”. For any exception or error, the server will respond with ServerResponse “failed” along with the reason for the error in the OptionalField[“ErrorMessage”].

Also, all the standard errors provided by gRPC are supported by default. More info on the topic can be found here.

Appendix A: Order Event Types

Event Types	Description
ClerkReject	Eze EMS rejects a user order
ExchangeAcceptOrder	Destination accepts event submitted
ExchangeBustOrder	Order that was busted by the Exchange
ExchangeKillOrder	Trade is busted by the destination
ExchangeReportStatus	Exchange reports the status of an order
ExchangeTradeOrder	Exchange executes a fill
ForeignExecution	Doneaway trade and allocation events • Allocation: Executed in another account on the EMS • Doneaway: Executed outside of the EMS but recorded in EMS
UserSubmitAllocation	User submitted allocation
UserSubmitAllocationEx	Outbound allocation
UserSubmitBust	Submission of a bust (trade or allocation) by a user
UserSubmitCancel	User submitted cancellation of a fill
UserSubmitChange	Change to an order by a user
UserSubmitCompoundOrder	Submission of a pair order by a user to an exchange
UserSubmitOrder	Submission of an order by a user to an exchange
UserSubmitStagedOrder	Submission of a staged order by user
UserSubmitTradeReport	User enters a trade through the trade report functionality (but does not mark it as a foreign execution)

Note: More information can be found in the **Reason** field in case of a Order rejection or failure.